

Question Answering in the Context of Stories

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In this study a model of question answering (called *QUEST*) is tested in the context of short stories. College students first read a story and then judged the quality of answers to questions about episodes in the story. The model could account for the goodness-of-answer judgments and decision latencies for 5 categories of questions: why, how, when, enablement, and consequence. *QUEST* specifies the information sources that are activated during question answering; the content of each information source is structured according to a theory of knowledge representation. *QUEST* specifies the convergence processes that dramatically narrow down the set of possible answers (activated from the information sources) to a small set of good answers to a question.

Question answering is a very important activity during knowledge acquisition, communication, and social interaction. When individuals learn about a new topic, they frequently ask experts questions to acquire critical, relevant information. Many speech acts are questions and responses to questions during normal conversations (D'Andrade & Wish, 1985). Question answering underlies nearly all experiments that are conducted on humans in laboratory settings. The subjects want to know *why* the experiment is being conducted, *how* they should respond to stimulus materials (which are often explicit questions), and *what the consequences* of the study are. In view of the intrinsic importance of question answering as a psychological phenomenon, the field would benefit from a well-developed theory of question answering.

The present study tests a psychological model of question answering called *QUEST*, which was recently developed by Graesser and Franklin (1990). *QUEST* handles a broad diversity of questions and knowledge domains. The types of questions include verification questions (e.g., is statement S true or false), concept completion questions (e.g., who, what, when, and where), and questions that elicit the causes, reasons, or consequences of actions (e.g., why, how, and what if). During the course of developing the *QUEST* model, question-answering protocols have been analyzed in the context of short stories (Graesser, 1981; Graesser & Clark, 1985; Graesser & Murachver, 1985; Graesser, Robertson, & Anderson, 1981), lengthy fairy tales (Graesser, Robertson, Lovelace, & Swinehart, 1980), scripts (Graesser, 1978), expository texts on scientific mechanisms (Graesser, 1981; Graesser & Hemphill, 1991; Graesser, Hemphill, & Brainerd, 1989), telephone sur-

veys, televised interviews, and tape-recorded business transactions (Graesser, Roberts, & Hackett-Renner, 1990).

A distinction is frequently made between closed-class questions and open-class questions. Appropriate replies to closed-class questions are restricted to a limited number of alternatives and are normally short answers. For example, answers to verification questions are "yes," "no," "maybe," or "I don't know." Psychologists have investigated closed-class question answering at considerable depth (Clark & Clark, 1977; Glucksburg & McCloskey, 1981; Reder, 1982, 1987; Singer, 1986, 1990). In contrast, appropriate responses to open-class questions are elaborate verbal descriptions. For example, answers to why-, how-, and what-if questions frequently contain several sentences of information. A scientific analysis of complex verbal descriptions is a difficult challenge, both methodologically and theoretically, but there has been some progress in understanding open-class questions in psychology (Collins, Warnock, Aiello, & Miller, 1975; Graesser & Black, 1985; Graesser & Golding, 1988; Hilton, 1990; Norman, 1973; Norman & Rumelhart, 1975; Piaget, 1952; Shanon, 1983; Trabasso, van den Broek, & Lui, 1988).

The *QUEST* model has incorporated a number of computational formalisms and insights from models of question answering in artificial intelligence and computational linguistics (Allen, 1987; Bruce, 1982; Dahlgren, 1988; Dillon, 1982; Kaplan, 1983; Lehnert, 1978; Lehnert, Dyer, Johnson, Young, & Harley, 1983; McKeown, 1985; Winston, 1984; Woods, 1977). Whenever formalisms and claims are borrowed from other fields, it is important to conduct psychological studies to assess whether they have relevance to human cognition. The present study investigates whether adults' judgments of answer quality can be explained by some of the components of *QUEST* that have foundations in artificial intelligence and computational linguistics.

This study focuses on five categories of open-class questions: why, how, when, what enabled X, and what are the consequences of X (abbreviated as CONS). After college students read a short narrative passage, they evaluated the quality of particular answers to particular questions about actions and events in the story plot. The answers to the questions were scaled on a number of variables that are relevant to *QUEST*'s components. We performed multiple

This research was funded by contracts awarded to Arthur C. Graesser by the Office of Naval Research (N00014-88-K-0110 and N00014-90-J-1492).

We would like to thank Karl Haberlandt, Deborah McClutchen, Tom Trabasso, and some anonymous reviewers for their comments on earlier versions of this article.

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regression analyses to determine whether these theoretical variables could predict the "goodness-of-answer" (GOA) judgments and decision latencies. In some conditions, the subjects furnished discrete, binary GOA judgments for each question-answer pair. That is, they decided whether an answer was a "good" answer versus a "bad" answer to a particular question. Decision latencies were recorded for these binary GOA judgments. In other conditions, subjects supplied GOA ratings on a four-point scale instead of binary GOA judgments.

The subsequent section briefly illustrates how the QUEST model would explain the quality of particular answers to questions about episodes in stories. The section provides a concrete example of an information source and shows how a convergence mechanism isolates a few relevant answers to a question, given that dozens to hundreds of information units are activated in working memory. The primary focus of this study is to test the convergence mechanism. After the QUEST model is illustrated with some concrete examples, we describe the components of QUEST in more detail and identify patterns of data that would be generated by each component of QUEST. We point out existing evidence for each component, to the extent that empirical data are available, and identify other models of question answering that have similar components. After the QUEST model has been presented, we report a study that tests QUEST's convergence mechanism.

Illustration of QUEST's Convergence Mechanism

Suppose that a person reads the story *The Czar and his daughters*, which is presented in Table 1. Suppose further that the knowledge structure in Figure 1 consists of the cognitive representation of the story plot. That is, the structure includes all of the explicit statements in the story, except for the first sentence, which announces the setting (i.e., "Once there was a Czar who had three lovely daughters."). The story structure consists of 25 *statement nodes* that are connected by 30 relational *arcs* (i.e., Reason, Consequence, Implies, Outcome, and Initiate).

QUEST predicts which of the nodes from the knowledge structure are good answers to a particular question. For example, suppose that the person is asked, "Why did the daughters stay in the woods too long?," a question that refers to Node 5 in Figure 1. There are 24 nodes in Figure 1 that are potential answers to this question (not counting Node 5). However, most of the nodes in Figure 1 would not serve as good answers to this question. The convergence mechanism identifies those nodes from a knowledge structure that are good answers to a particular question.

Consider the following possible answers to "Why did the daughters stay in the woods too long?":

- (A) Because the daughters forgot the time. (Node 4)
- (B) Because the dragon kidnapped the daughters. (Node 7)
- (C) Because the daughters were walking in the woods. (Node 2)
- (D) Because the heroes fought the dragon. (Node 18)

Although all four of the answers are stored in the knowledge structure, only one answer would be a very good answer to

Table 1

Passages Used in Experiments

The Czar and His Daughters

Once there was a Czar who had three lovely daughters. One day the three daughters went walking in the woods. They were enjoying themselves so much that they forgot the time and stayed too long. A dragon kidnapped the three daughters. As they were being dragged off they cried for help. Three heroes heard the cries and set off to rescue the daughters. The heroes came and fought the dragon and rescued the maidens. Then the heroes returned the daughters to their palace. When the Czar heard of the rescue, he rewarded the heroes.

The Ant and the Dove

A thirsty ant went to the river. He became carried away by the rush of the stream and was about to drown. A dove was sitting in a tree overhanging the water. The dove plucked a leaf and let it fall. The leaf fell into the stream close to the ant and the ant climbed onto it. The ant floated safely to the bank. Shortly afterwards, a birdcatcher came and laid a trap in the tree. The ant saw his plan and stung him in the foot. In pain, the birdcatcher threw down his trap. The noise made the bird fly away.

the question. Specifically, Answer A would survive all three components of QUEST's convergence mechanism: the arc search procedure, structural distance, and constraint satisfaction. The *arc search procedure* for a particular question category (e.g., why-event) selects those nodes that are on *legal* paths of arcs radiating from the queried node and prunes out those nodes that are on *illegal* paths. Answer B would not be good because it is not on a legal path of arcs between the queried node (Node 5) and the answer node (Node 7). Specifically, answers to why-event questions are not on paths that include forward I-arcs and O-arcs. According to the arc search procedure for why-event questions (which is specified later), Nodes 1-4 in Figure 1 are on legal paths that radiate from Node 5, whereas Nodes 6-25 are on illegal paths. In this example, only 17% of the nodes from Figure 1 (4 out of 24) would pass the arc search procedure.

The second component of convergence, *structural distance*, selects nodes from a knowledge structure that are close structurally to the queried node. That is, there is a negative correlation between goodness-of-answer and the number of arcs that connect the answer node to the queried node. Given that Event 5 is the queried node in the example question, the structural distance scores would be 1, 2, 3, and 7 arcs for Example Answers A, B, C, and D, respectively. The structural distance component would yield the following ordering on answer quality: $A > B > C > D$.

The third component of convergence, *constraint satisfaction*, prunes out a node from the knowledge structure if the semantic and conceptual content of the node is incompatible with the content of the queried node. For example, the answer should not contradict the queried node. Answer C passes the arc search procedure and is moderately closed to the Queried Node 5. However, Answer C does not have a strong causal relationship with Node 5, so it would not pass constraint satisfaction. That is, walking in the woods does not have a strong causal relationship with staying too long in the woods. QUEST considers several other dimensions for evaluating

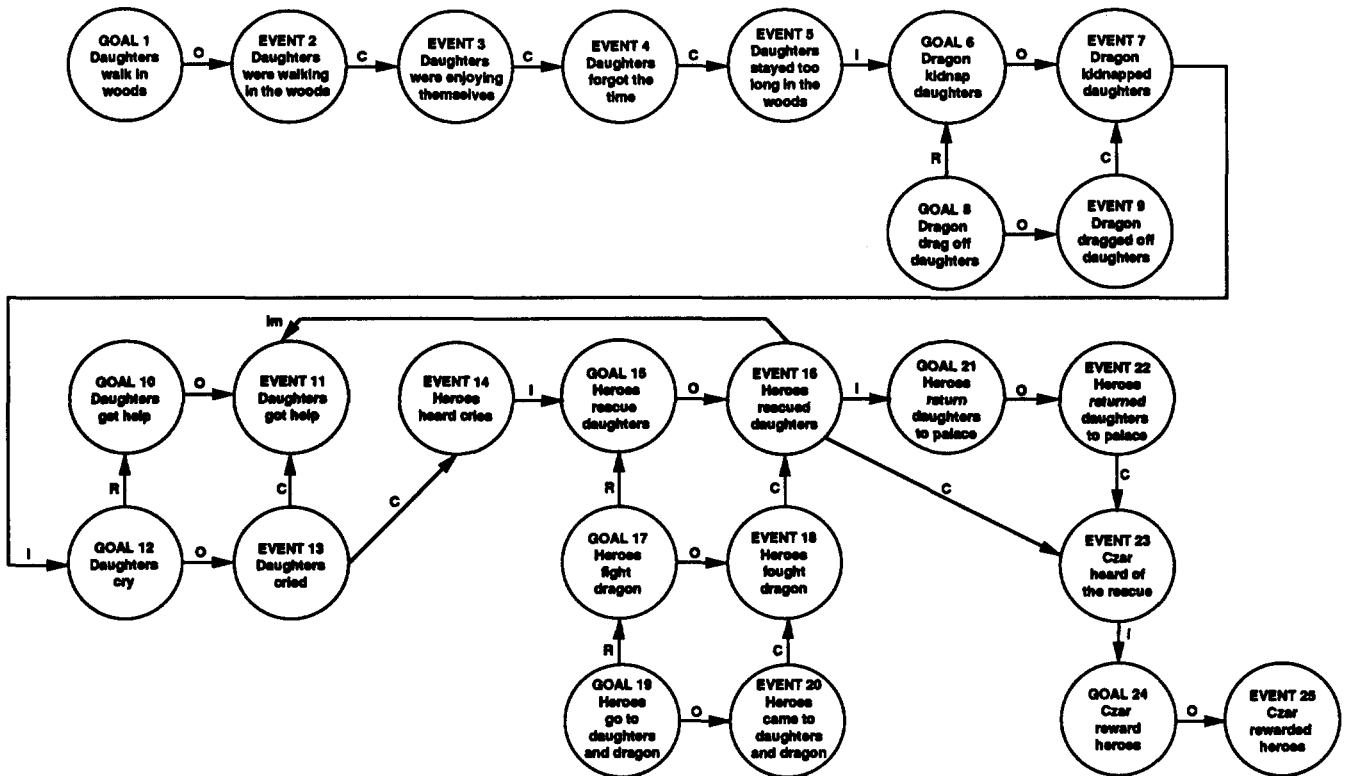


Figure 1. An example conceptual graph structure that captures the plot in *The Czar and his Daughters*. (The statement nodes are in the circles. The arc categories are Consequence [C], Reason [R], Initiate [I], Outcome [O], and Implies [Im].)

candidate answers on constraint satisfaction, as is discussed later.

Answer D ("because the heroes fought the dragon") would be a particularly bad answer to the question "Why did the daughters stay too long in the woods?" because it fails on all three components of convergence. First, Answer D is illegal; it would not be generated by QUEST's arc search procedure. Second, Answer D is not structurally close to the queried node because its structural distance score is 7. Third, there is little or no causal relationship between Answer D and the queried node. Answer quality is expected to decrease to the extent that an answer violates more components of convergence.

Only 2 out of the 24 nodes are good answers when all three of QUEST's convergence components are applied to the example question and Figure 1. The good answers are Node 4 ("because the daughters forgot the time") and Node 3 ("because the daughters were enjoying themselves"). Therefore, only 8% of the nodes in the knowledge structure end up surviving the three components of the convergence mechanism.

There is a fundamental distinction between *intentional actions* and *involuntary events* in the representational system adopted by QUEST. An intentional action is an amalgamation of (a) a goal node and (b) an event that achieves the goal by virtue of the agent executing some plan. For example, the intentional action *The heroes fought the dragon* is an amalgamation of the Goal Node 17 (Heroes wanted to fight the dragon.) and the Event Node 18 (Heroes fought the dragon.).

In contrast, involuntary events are not motivated by the goals of agents. Involuntary events are products of physical, cognitive, or social systems that are not governed by the goals and plans of agents. Examples of involuntary events are Nodes 4, 5, 14, and 23. According to QUEST, the arc search procedures are quite different for intentional actions (or *actions* for short) and involuntary events (or *events* for short).

The queried node is an intentional action in the question "Why did the heroes fight the dragon?" The arc search procedure for why-action questions includes superordinate goals (through forward Reason-arcs), but not subordinate goals (through backward Reason-arcs). Therefore, Answer E would be generated by QUEST, but not Answer F.

- (E) Because the heroes wanted to rescue the daughters. (Node 15)
- (F) Because the heroes wanted to go to the daughters. (Node 19)

In contrast, the arc search procedure for how-action questions includes subordinate achieved goals, but not superordinate achieved goals. Therefore, a good answer to "How did the heroes fight the dragon?" would be answer F', but not E'.

- (E') The heroes rescued the daughters. (Node 16)
- (F') The heroes went to the daughters. (Node 20)

Each question category has its own unique arc search procedure. Consequently, the good answers to questions are expected to vary among question categories.

This section has briefly illustrated how QUEST's convergence mechanisms narrow the "node space" from dozens of nodes in a single knowledge structure to a small number of good answers to a question. In most cases, however, there are several knowledge structures that serve as relevant information sources for a question. Moreover, there are other components of QUEST that are needed for an adequate explanation of question answering. The next section provides a more complete and detailed description of the QUEST model.

QUEST Model of Question Answering

It is convenient to segregate QUEST into four major parts: question categorization, information sources, the convergence mechanism, and pragmatic components. At the same time, however, we acknowledge that these major parts are highly integrated and interrelated (Dyer, 1983; Lehnert et al., 1983; Robertson, Black, & Lehnert, 1985). QUEST determines the relevant question category for a particular query during the process of *question categorization*. There is a catalog of question categories, each of which has a unique procedure that operates on knowledge structures and produces replies. The *information sources* for a question consist of the set of world knowledge structures that are relevant to a question and that provide the answer content. According to QUEST, each information source is a structured entity that contains a large set of nodes and arcs. Obviously, most of the nodes from the various information sources are not produced when a particular question is answered. The *convergence mechanism* narrows down the node space to a few good answers to a question. Regarding *pragmatics*, QUEST has components that address the goals and the shared knowledge of the questioner and answerer.

QUEST was not developed to account for the linguistic features of questioning. QUEST does not explain the process of parsing the question syntactically and the process of articulating replies linguistically. Instead, QUEST was developed to account for the conceptual content of the answers.

Information Sources

Each information source is a structured database that contains a set of statement nodes that are interrelated by directed arcs (e.g., Figure 1). QUEST distinguishes between specific and generic knowledge structures. A *specific knowledge structure* represents a particular text or experience that is encoded at a particular time and place. From the standpoint of the present study, the important specific knowledge structure is the *textbase* that is constructed when the reader comprehends a particular text (Kintsch, 1988; Kintsch & van Dijk, 1978).

A *generic knowledge structure* is a summary representation for a set of specific knowledge structures. Generic knowledge structures include schemata (Rumelhart & Ortony, 1977), scripts (Schank & Abelson, 1977), frames (Minsky, 1975), memory organization packages (Schank, 1982), thematic affect units (Dyer, 1983), concepts (Bourne, 1966; Smith & Medin, 1981; Wyer & Gordon, 1984), and other constructs that are designed to represent sets of exemplars. These various constructs have distinctive formal properties and content, but

the version of QUEST in this article does not capitalize on these distinctions and does not make strong assumptions about the packaging of information in long-term memory. It suffices to say that a generic knowledge structure is associated with each content word in a question. For example, the generic knowledge structures in the question "Why did the daughters cry?" are *daughter* and *cry*. If this question was asked after a person read the Czar story in Table 1, then the information sources would include the textbase (i.e., Figure 1) and two generic knowledge structures (daughter, cry).

Researchers do not agree on the extent to which inferences are incorporated into the textbase. At one extreme, there is the "explicit textbase" position that assumes that the textbase contains primarily, if not exclusively, those nodes that are explicitly mentioned in the text (Kintsch & van Dijk, 1978). At the other extreme, there is a "massive elaborative inference" position that assumes that a large number of knowledge-based inferences are activated during comprehension and incorporated into the textbase. There are a variety of intermediate positions that declare that particular subsets of inferences are incorporated into the textbase during comprehension (Graesser & Bower, 1990). These models of inference processing have been tested in experiments that can assess whether particular classes of inferences are generated "on-line" during comprehension. Some models assume that inferences are constructed during comprehension if they are easy to generate or if they are needed to establish text coherence, whereas most elaborative inferences are not incorporated in the textbase (Graesser & Clark, 1985; McKoon & Ratcliff, 1989; Potts, Keenan, & Golding, 1988). Coherence-based inferences bridge the conceptual gaps that must be filled to interpret and connect the explicit clauses in the text. Elaborative inferences merely embellish the explicit textbase; they are not needed to construct a coherent textbase. A model by Long, Golding, Graesser, and Clark (1990) assumes that elaborative inferences are automatically generated if they help explain *why* an actor or event in a story plot occurs; elaborative inferences are not incorporated into the textbase if the inferences elaborate *how* the action or event occurs or if they are expectations about subsequent episodes in the plot (i.e., answers to what-happened-next questions). The present study provides indirect tests of these positions by assessing the relationship between structural distance and goodness-of-answer judgments. More specifically, the alternative positions provide different structural distance scores (i.e., the number of arcs in the textbase between the queried node and the answer node). The most plausible position should produce the highest negative correlation between structural distance and GOA judgments.

Whenever a question taps a particular information source, there is an "entry node" in the information source that matches the queried node. For example, consider the question "Why did the daughters cry?" which has the queried node "daughters cried." The entry node in the textbase would be the explicit node "the daughters cried" (Node 13), a perfect match to the queried node. Sometimes there are partial matches between the queried node and the entry node. This frequently occurs when generic knowledge structures are tapped for answers. For example, the entry node in the generic knowledge structure for *cry* would be "X cries." This entry

node matches the queried node (daughters cried), when X is substituted for daughters.

Very often there are nodes from two different knowledge structures that end up matching. For example, "daughter is weak" in the *daughter* structure partially matches "X is weak" in the *cry* structure. When considering all of the information sources associated with a question, there is a set of "intersecting nodes" that includes all matching nodes between two or more information sources. These intersecting nodes have a privileged status because they tend to be produced as answers more often than nonintersecting nodes, when individuals answer questions about actions and events in stories (Graesser & Clark, 1985). The present study investigates whether GOA judgments are higher for those answers that intersect among the textbase and the generic knowledge structures associated with the content words in the question.

There is some debate over the relative contributions of the textbase and the generic knowledge structures when questions are answered. According to Reder's (1987) model of question answering, individuals can strategically tap either the textbase or the generic knowledge when they decide whether a particular sentence is true or false (i.e., verification judgments) and when they decide whether a test sentence had been presented earlier (i.e., recognition judgments). The person can either access a specific memory (corresponding to a textbase) or rely on plausibility judgments (derived from the generic knowledge structures) when these verification judgments and recognition judgments are made. As the delay between text comprehension and question answering increases, individuals rely more on generic knowledge, because the textbase is less accessible from memory (Graesser & Clark, 1985; Graesser & Nakamura, 1982; Kintsch, 1988).

In some instances, an answer to a question comes exclusively from a textbase because there are no generic knowledge structures that would furnish the information. This occurs when the reader comprehends an expository text on an unfamiliar topic. At the other extreme, an answer could be extracted from several generic knowledge structures in addition to the textbase. This occurs when a reader comprehends a simple story with a prototypical plot and familiar content words. It is possible to scale each answer on the number of generic information sources that furnish that answer to a question. Interestingly, there may not be a linear relationship between GOA judgments and the number of generic information sources producing an answer. There may be a trade-off between the accessibility of an answer and the informativeness of an answer. Those answers that come from multiple information sources have the advantage of being more accessible from memory; yet they are more likely to be in the common ground of speech participants and thereby less informative.

Conceptual Graph Structures

Each information source is represented in the form of a conceptual graph structure. A conceptual graph structure consists of a set of categorized statement nodes that are interrelated by a set of categorized, directed arcs. Conceptual graph structures have a foundation in a number of represen-

tational systems in the cognitive sciences, such as propositional theories (Anderson, 1976; Clark & Clark, 1977; Kintsch, 1974; Norman & Rumelhart, 1975), story grammars (Mandler, 1984; Stein & Glenn, 1979), causal chain theories (Trabasso & van den Broek, 1985; Trabasso et al., 1988; Trabasso, van den Broek, & Suh, 1989), conceptual dependency theory (Alterman, 1988; Schank & Abelson, 1977; Schank & Riesbeck, 1981), theories of rhetorical organization (Mann & Thompson, 1986; Meyer, 1985), and Sowa's (1983) conceptual graph representations. It is beyond the scope of this article to clarify the exact roots and the utility of conceptual graph structures (see Graesser & Clark, 1985; Graesser & Franklin, 1990; Graesser & Goodman, 1985). Instead, we provide a satisfactory description for the purpose of this article.

Figure 1 presents a conceptual graph structure for the explicit actions and events in the plot of the Czar story. The circles contain statement nodes that are analogous to propositions. A statement node contains a predicate (e.g., verb, adjective, and adverb) and one or more arguments (e.g., noun and embedded proposition). Each argument satisfies a particular role (e.g., agent, object, recipient, time, and location). However, a statement node differs from a proposition unit in two ways. First, a statement node contains two or more propositions whenever a statement node has an embedded proposition. For example, "The Czar heard about the heroes rescuing the daughters" is a single statement node, but has two propositions: PROP1 (hear-about, Czar, PROP2) and PROP2 (rescue, heroes, daughters). Second, unlike propositions, each statement node is assigned to a node category (i.e., state, event, goal, and style). *State* nodes are ongoing attributes of the world that remain unchanged from the perspective of the relevant time frame (i.e., the story plot). Examples of state nodes are "The czar had three daughters" and "The daughters are lovely." *Event* nodes involve a state change at some point within the relevant time frame (see event nodes in Figure 1). *Goal* nodes depict a state or event that is desired by an agent. A *style* node conveys the intensity, speed, or qualitative pattern in which an event or action occurs (e.g., X occurred quickly, in circles).

As discussed earlier, there is an important distinction between intentional actions and involuntary events. An intentional action is always inspired by an agent's goals. In QUEST's representational system, an action is an amalgamation of a goal node and an event, state, or style node that signifies the successful achievement of the goal. The action "the hero killed the villain" is an amalgamation of "the hero wanted to kill the villain" and "the villain was killed." Given that goals and outcomes are distinguished in most theories of motivation (Sorrentino & Higgins, 1986) and in theories of social cognition that map out knowledge structures (Pennington & Hastie, 1986; Read, 1987), the two aspects of an action were explicitly segregated in QUEST's representational system.

The statement nodes are interrelated by categorized, directed arcs. Each arc is directed in the sense that the arc radiates from a "source node" and feeds into an "end node." This directedness implies that there is an asymmetrical relation between the nodes. There are nine arc categories in

QUEST, but only six need to be identified in this article: Consequence (C), Implies (Im), Manner (M), Initiate (I), Reason (R), and Outcome (O). Associated with each arc category are conceptual criteria and rules of composition that constrain when and how nodes are connected by the particular category. Table 2 presents a description of each arc category, along with its temporal constraints, its rules of composition, and an example. A more extensive description of the arc categories is reported in previous publications (Graesser & Clark, 1985; Graesser & Franklin, 1990; Graesser & Murachver, 1985).

Convergence Mechanism

According to QUEST, several information sources are activated when a question is asked and each information source has a large number of nodes. The convergence mechanism narrows down the node space to a comparatively small number of good answers to a question. Convergence is accomplished by (a) the arc search procedure associated with the question category, (b) structural distance, and (c) constraint satisfaction.

Arc search procedure. The arc search procedure determines the legal paths of arcs that radiate from the entry node in an information source. For example, consider the question "What enabled the heroes to hear the cries?" The entry node is Event 14 in Figure 1, and the question category is "enable-event." The *legal* answers generated by this question category are causal antecedents to the entry node. Causal antecedents consist of all nodes that radiate from the entry node on paths that have any combination or sequence of the following arcs: backward C, forward or backward Im, backward O, and backward I. The legal answers to "What enabled the heroes to hear the cries?" are Nodes 1-9, 12, and 13; the illegal answers are Nodes 10, 11, and 15-25. Therefore, the arc search procedure defines which nodes in the structure are legal answers and which are illegal answers. It should be noted that the inverse of a causal antecedent path is a causal consequence path: any combination or sequence of forward C, forward or backward Im, forward O, and forward I arcs.

Arc search procedures have been identified for why-, when-, enable-, and consequence (CONS)-questions. The validity of these arc search procedures has been supported in previous psychological studies. QUEST's arc search procedures have significantly predicted the likelihood of nodes being produced as answers to the five types of questions (Graesser & Hemphill, 1991; Graesser et al., 1989; Graesser & Murachver, 1985) and the ability of subjects to discriminate good answers from bad answers (Golding, Graesser, & Millis, 1990; Murachver, Murray, & Graesser, 1985). The arc search procedures have consistently been robust predictors of both the answer production scores and the GOA judgments. This gives some credence to those models of question answering in artificial intelligence and computational linguistics that have emphasized the importance of knowledge organization and of restricting search by pursuing particular conceptual relations (Dahlgren, 1988; Lehnert, 1978; Lehnert et al., 1983; Souther, Acker, Lester, & Porter, 1989; Winston, 1984).

Table 2
Six Categories of Arc

Arc category	Description and temporal constraints	Composition rule	Example
Consequence (C)	A causes or enables B	(event state style) -C→	(Event: The daughters cried) -C→
Implies (Im)	A precedes B in time A implies B	(event state style)	(Event: The heroes heard the cries)
Reason (R)	A and B overlap in time B is a reason or motive for A B is a superordinate goal of A	(event state style) -Im→ (event state style) (goal) -R→ (goal)	(State: The heroes were strong) -Im→ (State: The heroes were powerful) (Goal: Heroes wanted to fight dragon) -R→
Outcome (O)	A is achieved before B is achieved B specifies whether or not the goal A is achieved	(goal) -O→ (event state style)	(Goal: Heroes wanted to rescue the daughters) -O→ (Event: The daughters were rescued)
Initiate (I)	A precedes B in time A initiates or triggers the goal in B	(event state style) -I→ (goal)	(State: The daughters were in trouble) -I→ (Goal: Heroes wanted to rescue the daughters)
Manner (M)	A precedes B in time B specifies the manner in which A occurs A and B overlap in time	(goal) -M→ (goal style) (style) -M→ (style) (event) -M→ (event style)	(Goal: Heroes want to go to dragon) -M→ (Goal: Heroes want to run to dragon) (Event: The daughters cried) -M→ (Style: The cries were loud)

The arc search procedures are quite different for involuntary events and intentional actions. In the case of events, answers to why-, how-, enable-, and when-questions include causal antecedents to the queried node, whereas answers to CONS-questions pursue causal consequences (and some causal antecedents of causal consequences). However, there are some additional differences among question categories, as specified below.

<i>why-event.</i>	Causal antecedents of queried event.
<i>how-event.</i>	Causal antecedents of queried event, style specifications (through forward M-arc), or both.
<i>enable-event.</i>	Causal antecedents of queried event.
<i>when-event.</i>	Causal antecedents of queried event, time indices (e.g., in the morning, 4 days ago), or both.
<i>CONS-event.</i>	Causal consequences of queried event, causal antecedents of causal consequences, or both.

It should be noted that the why-event and enable-event questions have equivalent arc search procedures.

Queried actions have an additional level of complexity because actions are embedded in local goal hierarchies (Goldman, 1985; Graesser & Clark, 1985; Trabasso et al., 1988; Voss, Greene, Post, & Penner, 1983). A local goal hierarchy captures the plan that an agent desires to achieve by virtue of executing the action. Suppose that the queried action is "The heroes fought the dragon" (an amalgamation of Goal 17 and Event 18 in Figure 1). The agent in this example is the *heroes* and the local goal hierarchy includes Nodes 15, 17, and 19. This local goal hierarchy is different from the goal hierarchies of other agents (e.g., dragon and daughters) and different from the heroes' goal structures at other points in the plot. The goal-action nodes in a legal goal hierarchy may be segregated into superordinate nodes (which are accessed by pursuing paths of forward R and backward M arcs) versus subordinate nodes (accessed through backward R and forward M arcs). Moreover, there is an important distinction between the goal form and the action form of each goal-action node. For example, "The heroes wanted to fight the dragon" is a goal form, whereas "The heroes fought the dragon" is an action form.

The legal answers and answer forms are presented below for the five question categories with respect to the local goal hierarchy).

<i>why-action.</i>	Superordinate goals.
<i>how-action.</i>	Subordinate actions.
<i>when-action.</i>	Superordinate goals-actions and subordinate actions.
<i>enable-action.</i>	Superordinate goals and subordinate actions.
<i>CONS-action.</i>	Superordinate actions.

The legal answers to queried actions are not confined to the local goal hierarchy. Legal answers to queried actions also include the paths specified by the arc search procedures of

event questions. That is, the arc search procedures that were specified earlier for the queried events are also applied to (a) the queried actions and (b) the set of legal answers from the local goal hierarchy. For example, legal answers to "Why did the heroes fight the dragon?" include the superordinate goal (Node 15), the causal antecedents to the queried action, and the causal antecedents of the superordinate goal (i.e., Nodes 1-9 and 12-14). Subordinate actions (Nodes 19-20) are illegal answers to why-action questions.

The arc search procedures of QUEST have been implemented in a computer program written in LISP. The program operates on conceptual graph structures that the user builds. To ask a question, the user specifies the question category (why, how, when, enable, and CONS) and the node in the information source to query. The program prints out the legal answers to the question, following QUEST's arc search procedures.

Structural distance. The nodes in an information source have a lower likelihood of being good answers to a question to the extent that the nodes are further from the entry node. Therefore, GOA judgments should be negatively correlated with the number of arcs between the entry node (corresponding to the queried node) and the answer node. Empirical support for this negative relationship has been found for narrative text (Golding et al., 1990; Murachver et al., 1985), for expository text (Graesser & Hemphill, 1991; Graesser et al., 1989), and for generic knowledge structures (Graesser & Clark, 1985). In these studies, the effects of the arc search procedures were partialled out from the effects of structure distance. Trabasso et al. (1989) also reported that the distance between two clauses in a causal structure predicts the strength of their causal relationship, even when plausible extraneous variables are partialled out.

The predicted effects of structural distance on GOA judgments would be generated by some models of question answering other than QUEST (Shanon, 1983; Winston, 1984). For example, Winston's model simulates answers to why-questions and how-questions in the context of goal-oriented structures. His question-answering algorithms specify that good answers are only one arc from the queried node. Theories of marker passing and spreading activation (Anderson, 1976, 1983) would also furnish such a prediction.

As discussed earlier, the structural distance between question and answer depends on the number of knowledge-based inferences that are incorporated into the textbase. The distance scores increase as a function of the number of inferences in the textbase. Unlike the previous studies that have tested QUEST, the present study tests alternative positions regarding the number of inferences in the textbase, as opposed to inferences that reside entirely in the generic knowledge structures associated with the text. Position 1 assumes there are no knowledge-based inferences in the textbase; all inferences are in the generic knowledge structures. Position 2 assumes that the textbase contains all inferences that would ever be manifested as answers to questions. Position 3 assumes that some inferences are in the textbase, namely those that would surface as answers to why-questions. There is some evidence that answers to why-questions tend to elicit inferences that are generated during comprehension, whereas answers to how- and CONS-questions tend not to include comprehension-

generated inferences (Graesser, Haberlandt, & Koizumi, 1987; Long et al., 1990). Once again, each of these positions yields different structural distance scores. Support for a position consists of a comparatively high negative correlation between structural distance and GOA judgments.

Constraint satisfaction. A good answer must satisfy semantic and conceptual constraints that are propagated by the queried node. For example, the answer node must not contradict the queried node. The constraint satisfaction component in QUEST eliminates potential answer nodes that are conceptually incompatible with the queried node. Graesser and Clark (1985) identified some conceptual constraints that were needed to explain convergence when inferences are generated during narrative comprehension. Graesser and Clark's constraints were adopted in the present study, along with one additional constraint that involves causality. It should be noted that the constraint satisfaction component has not previously been tested in the context of question answering and story passages.

We computed one overall constraint satisfaction measure that varied from 0 (*answer does not satisfy any constraints*) to 1 (*answer satisfies all constraints*). This overall measure was based on five dimensions: argument overlap, temporal compatibility, planning compatibility, plausibility (truth), and causal strength. Each dimension had values ranging from 0 to 1. The overall constraint satisfaction score for a question-answer item was an average of the five dimensions.

1. *Argument overlap.* This dimension indicates whether any of the noun arguments in the answer node fail to match an argument in the queried node. A high score means that the answer has no new argument nouns, whereas a low score means that the answer has one or more new argument nouns. For example, consider the queried node "The dragon kidnapped the daughters." The answer "the dragon hated the daughters" receives the value of 1 because both of its arguments (dragon and daughters) are arguments in the queried node. The answer "the heroes loved the daughters" receives the value of 0 because it has a new argument (heroes) that is not in the queried node. Sometimes a new argument in an answer was considered only minor (the word *time* in "the daughters forgot the time"); these cases had a value of .5 instead of 0. We defined major new arguments as story characters, whereas all other new arguments were regarded as minor. Argument overlap may be construed as an index of referential coherence between the answer and the question (Kintsch & van Dijk, 1978; Trabasso et al., 1989). When the major characters in the answer are all referenced by arguments in the question, then the answer is more coherently related to the question than when the answer introduces a new character.

2. *Temporal compatibility.* This dimension specifies whether the answer occurs within the time span of the queried event or action. A time span is associated with the queried node, from the perspective of the plot chronology. An answer has 0 temporal compatibility when it falls outside of the scope of this time span. If the queried node were "The dragon kidnapped the daughters," then the answer "the daughters were born" falls entirely outside of the time span (value = 0) and the answer "the daughters were frightened" falls within the time span (value = 1). A value of .5 is reserved for

borderline cases. It is possible to define temporal overlap more precisely by developing a calculus for time intervals and durations (Allen, 1984). However, intuitive judgments by trained judges were adequate for the present purposes.

3. *Planning compatibility.* This dimension specifies whether the existence or occurrence of the answer node would block or seriously interfere with the plan underlying a queried action. This dimension therefore applies to queried actions, but not to queried events. If the queried node were "The dragon dragged off the daughters," the answer "the heroes ran to the daughters" would not interfere with the dragon's plan (value = 1), whereas "the heroes killed the dragon" would block the dragon's plan (value = 0). Borderline cases receive a value of .5. This dimension has theoretical foundations in plan-based theories of narrative comprehension (Dyer, 1983; Wilensky, 1983).

4. *Plausibility.* This dimension specifies whether the answer is a true statement in the context of the story. False statements and answers that contradict the queried node receive a value of 0, whereas true statements receive a value of 1; items receive a value of .5 when the truth value is indeterminate. An example of an implausible node in the context of the Czar story is "the dragon wanted to get ransom"; this is an extreme form of anthropomorphizing.

5. *Causal strength.* This dimension measures the extent to which the answer node is causally related to the queried node. We adopted Trabasso and van den Broek's system for determining the extent to which two statements in a text are causally related (Trabasso & van den Broek, 1985; Trabasso et al., 1988, 1989; van den Broek, 1990). The strength of causality is computed on the basis of four subdimensions: temporality, operativity, necessity, and sufficiency. Suppose that the relationship "X cause Y" is being evaluated. *Temporality* is satisfied and receives a value of 1 if X precedes Y in time. *Operativity* is satisfied if X exists or occurs at the time when Y occurs. Stated differently, X is operating when Y occurs. *Necessity* is satisfied if X is a necessary condition for Y in the context of the story. Necessity receives a value of 1 if a counterfactual test is satisfied; that is, if the negation of X results in the negation of Y, then X is necessary for Y. It should be noted that this sense of necessity does not mean that X necessarily leads to Y or that X always leads to Y. *Sufficiency* is satisfied if X normally or frequently leads to Y. Suppose that X occurs and that the world continues in a normal fashion; if Y would occur under these conditions, then X is sufficient for Y.

For illustration, suppose that X is "the daughters cried" and Y is "the heroes fought the dragon." The value for temporality is 1 because the daughters' crying preceded the heroes' fighting the dragon. The value for operativity is an intermediate value of .5. It is unclear whether the daughters were still crying or were finished crying when the heroes fought the dragon. The value for necessity is clearly 1; if the daughters never cried, the heroes would have never been alerted to rescue them. The value of sufficiency is a borderline .5; it is unclear whether crying would or would not normally lead to someone fighting the dragon and rescuing the daughters.

There is a mathematical formula that expresses the overall causal strength between two events (X and Y), on the basis

of the values of temporality (T), operativity (O), necessity (N), and sufficiency (S). Specifically, causal strength is computed as $T * O * (N + S) / 2$. This mathematical formula has been reported in the psychological literature that investigates causal judgments (Einhorn & Hogarth, 1986) and the causal relatedness of events conveyed in text (van den Broek, 1990). According to this formula, the causal strength between the two example events is .38 [$1 \times .5 \times (1 + 5) / 2$]. The essence of this formula becomes clear when values of 0 versus 1 are substituted for T, O, N, and S. Temporality and operativity are both essential criteria for causality. In contrast, necessity and sufficiency are optional criteria in the sense that only one of the two criteria need to be met.

The causal strength between the queried node (Q) and the answer node (A) is quite different for the relation "A cause Q" and "Q cause A." When causal strength was scored for a particular Q-A pair, the selection of "A cause Q" versus "Q cause A" depended on the question category and its arc search procedure. For why-, how-, enable-, and when-questions, the causal strength of "A cause Q" was relevant because the theoretical answers to these questions consisted of causal antecedents. For CONS-questions, however, the relation "Q cause A" was relevant because these questions tap causal consequences. It follows that causal strength should be correlated with the arc search procedure (Graesser & Hemphill, 1991; Trabasso et al., 1989). In spite of this expected positive correlation, the causal strength and arc search components are not perfectly redundant. As illustrated in the previous section, an answer may pass the arc search procedure, but not be causally related to the answer. Multiple regression analyses permitted us to assess whether the arc search procedure and causal strength have significant independent contributions to GOA judgments.

Causal strength should also be negatively correlated with structural distance (Trabasso et al., 1989). The causal strength between Q and A should decrease as a function of the number of arcs between the Q and A.

Interactions Among Components of the Convergence Mechanism

QUEST specifies the three major components of the convergence mechanism, but does not make specific commitments about how the components interact during processing and as a result of processing. Moreover, there is no available research that sheds light on how the different components of convergence interact. One position is that arc search procedure, structural distance, and constraint satisfaction operate independently and additively. In this case, a decision is made after all three components are finished being executed. According to this position, there should be main effects, but no interactions, among the three components of convergence when multiple regression analyses are performed on GOA judgments and latencies. An entirely different position is that the components are highly interactive and interdependent. However, this position is not particularly informative unless it generates systematic output. In between these two extremes,

there are dozens of possible ways that the three components interact. The present study explores how the three components interact when GOA judgments are made.

It is possible that the components are executed in sequential order, such that the operation of Component N is dependent on the output from Component N - 1. For example, consider the processing sequence below.

- (1) Stage 1. Find the queried node and the answer node in the information source.
- (2) Stage 2. Apply the arc search procedure to determine whether the answer is on a legal path radiating from the answer node. If the answer is illegal, then stop and say "bad answer."
- (3) Stage 3. Apply the constraint satisfaction component to check whether the answer node is compatible conceptually with the question node. If the answer is not compatible, then stop and say "bad answer."
- (4) Stage 4. Apply the structural distance component. If the answer is structurally distant, then say "bad" answer. If the answer is structurally close to the queried node, then say "good answer."

Suppose this four-stage sequential model is applied to the question "Why did the daughters stay too long in the forest?" and to the alternative answers that were presented earlier. The various answers pass through differing numbers of stages. The theoretically good answer "because the daughters forgot the time" passes through all four stages (1-2-3-4). The theoretically bad answer "because the dragon kidnapped the daughters" fails both the arc search procedure and constraint satisfaction. Because it fails the arc search procedure, it passes through only two stages (1-2). It should be noted that an answer always passes through only two stages (1 and 2) if it fails the arc search component, regardless of its status on the constraint satisfaction and structural distance components. The answer "because the daughters were walking in the woods" passes arc search, but fails the constraint satisfaction component so it passes through three stages (1-2-3); its status on the structural distance component has no impact on answer quality. Any answer that passes both arc search and constraint satisfaction, but fails on structural distance, would pass through all four stages (1-2-3-4).

The above sequential processing mechanism predicts some interactions. First, there should be a two-way interaction between arc search and constraint satisfaction. Constraint satisfaction should predict GOA judgments for legal answers, but not for illegal answers (i.e., those that fail arc search); constraint satisfaction is not computed when the answer is illegal, according to the above sequential model. Second, there should be a three-way interaction among arc search, constraint satisfaction, and structural distance. That is, structural distance should influence GOA judgments for legal answers that satisfy constraints, but not for other types of answers. Third, decision latencies should be longer for legal answers than for illegal answers, with patterns of interactions analogous to those for GOA judgments.

Once again, the QUEST model does not dictate how the components of the convergence mechanism interact. The available empirical research and psychological models provide no foundation for generating predictions. The present study

is the first to explore interactions among the arc search, structural distance, and constraint satisfaction components.

Pragmatics

The pragmatic components address the social and communicative functions of answering a question. One component considers the *goals* of the questioner and answerer. From the perspective of the questioner, a question may be asked to acquire information, solve a problem, assess how much the answerer knows, persuade, control a conversation, and so on. From the perspective of the answerer, the answer may be formulated to inform the questioner, let the questioner know that the answerer knows something, entertain the questioner, and so on. A complete model of question answering would identify the goals of the speech participants in a particular discourse context and would determine how answers are tailored to achieve these goals (Allen, 1987; Appelt, 1985; Bruce, 1982; Clark, 1979; Clark & Schaefer, 1989; Francik & Clark, 1985; Graesser, Person, & Huber, in press; Kaplan, 1983). However, the present study does not test or manipulate the goal configurations of the questioner and answerer.

A second pragmatic component is the common ground (i.e., shared knowledge and mutual knowledge) between questioner and answerer (Clark & Marshall, 1981; Clark & Schaefer, 1989; Clark, Schreuder, & Butrick, 1983; Shanon, 1983). Common ground could have some surprising effects on answer selection and GOA judgments. If the common ground is high and the answerer wants to be informative (i.e., supplying information that the questioner might not know), then the answerer would avoid nodes that are explicitly mentioned in the passage structure and nodes that are stored in multiple information sources. There would be a negative correlation between GOA judgments and the number of generic information sources that supply an answer to a question. Regarding structural distance and common ground, there may be a preference for distant nodes because the proximate nodes tend to be obvious. If this were the case, there would be a positive correlation, rather than a negative correlation, between structural distance and GOA judgments.

According to some models (Clark & Marshall, 1981; Shanon, 1983), the answerer first evaluates the common ground between speech participants and then selects an answer that moderately extends the boundaries of the common ground. That is, the answer should be somewhat more informative, elaborate, or detailed than the common ground, but the answer should *not* be (a) entirely within the sphere of the common ground or (b) substantially more detailed than the common ground. These models predict an inverted-U function when GOA judgments are plotted as a function of the number of generic information sources and as a function of structural distance.

Overview of Tests of the QUEST model

This study investigated whether GOA judgments and decision latencies could be predicted by components of the QUEST model of question answering. College students first

read the narrative passages in Table 1 and later provided GOA judgments on question-answer pairs. There were five question categories (why, how, when, enable, and CONS), with half the questions about involuntary events and half about intentional actions in the stories. Some subjects provided GOA judgments on a discrete, binary scale ("good" vs. "bad" answer); decision latencies were collected for these judgments. Other subjects provided GOA ratings on a four-point scale.

A multiple regression approach was adopted in our tests of whether the GOA judgments and decision latencies could be predicted by QUEST's components. We were particularly interested in the three convergence components (arc search, structural distance, and constraint satisfaction) and the number of generic information sources that supply an answer to a question. Assuming that pragmatic components do not play a major role in this experiment, QUEST predicts positive beta-weights for the arc search, constraint satisfaction, and number of information sources, whereas a negative beta-weight is predicted for structural distance. Extraneous variables were controlled in the multiple regression analyses. The multiple regression approach permitted us to analyze two-way and three-way interactions among predictor variables, to explore how the components of convergence interact during processing. Curvilinear relationships between predictors and the dependent measures were also tested with the multiple regression analyses.

The present study tested several components and predictions of the QUEST model that have not been addressed in previous investigations of GOA judgments (Golding et al., 1990; Murachver et al., 1985). First, we tested whether constraint satisfaction predicts GOA judgments. Second, we explored interactions among the three components of convergence (arc search, structural distance, and constraint satisfaction). Third, we compared three measures of structural distance to test different hypotheses about which knowledge-based inferences are directly stored in the textbase. Fourth, we tested whether GOA judgments are influenced by the number of generic information sources that supply a particular answer to a question. Fifth, we assessed whether an inverted-U function characterizes the relationship between GOA judgments and structural distance and the relationship between GOA judgments and number of generic information sources that supply an answer to a question; these complex functions might occur if pragmatic components influence GOA judgments. Sixth, we explored whether GOA decision latencies can be predicted by components of QUEST.

Method

Subjects

The subjects were 110 undergraduate students at Memphis State University who participated in this experiment to fulfill a course requirement. Fifty of the subjects furnished binary GOA judgments, whereas the other 60 subjects furnished GOA ratings. Within each of these groups, an equal number of subjects was randomly assigned to one of the five question categories (why, how, when, enable, and CONS).

Materials

Table 1 presents the two stories that subjects read before judging the answers to the question-answer items. For each passage, 10 explicit nodes were selected as the queried nodes, yielding a total of 20 queried nodes. Half of the queried nodes were intentional actions and half were involuntary events. Because of a computer programming error, one of the queried events in each passage was omitted when the GOA ratings were collected. Therefore, there were 18 queried nodes in the GOA rating groups and 20 queried nodes in the discrete GOA judgment groups. The actions selected from the Czar story were as follows: The daughters walked in the woods, the dragon kidnapped the daughters, the dragon dragged off the daughters, the heroes fought the dragon, and the heroes returned the daughters to the palace. The events selected from the Czar story were as follows: The daughters enjoyed themselves, the daughters forgot the time, the daughters stayed too long, the heroes heard the cries, and the Czar heard about the rescue. The actions selected from the Dove story were as follows: The ant went to a river, the dove plucked a leaf, the ant got to shore, the birdcatcher put the trap in the tree, and the ant stung the birdcatcher on the foot. The events selected from the Dove story were as follows: The stream carried the ant away, the ant was about to drown, the leaf fell into the stream, the ant saw the birdcatcher's plan, and the birdcatcher threw down the trap.

Sixteen answer nodes were associated with each of the 20 queried nodes, which yielded a total of 320 question-answer items. These same 320 items were collected for each question category, except that the wording of the questions and answers varied somewhat among the categories. For example, if the queried node was "The heroes fought the dragon," then the five questions would be as follows: Why did the heroes fight the dragon?, How did the heroes fight the dragon?, When did the heroes fight the dragon?, What enabled the heroes to fight the dragon?, and What were the consequences of the heroes fighting the dragon?

Most of the answers to the questions were inferences (82%) as opposed to explicit statements in the passages. We restricted the number of explicit passage statements to 4 or fewer answers when the 16 answers were selected for each question. The inferences were sampled from question-answering protocols collected by Graesser and Murachver (1985). In that study, question-answering protocols were collected for each explicit statement node in the two stories. Each passage statement was probed with the five question categories (why, how, when, enable, and CONS); eight to 10 subjects generated answers to each of the five question categories. An answer distribution was associated with each particular question (e.g., "Why did the daughters cry?"). The answer distribution included all statement nodes that were produced as answers by 2 or more of the 8-10 subjects. An answer production score was computed for each unique answer, corresponding to the proportion of subjects who gave a particular answer to a particular question. All of the answers in the present study were selected from the answer distributions collected by Graesser and Murachver (1985).

Our method of sampling nodes from the answer distributions was stratified and random. A number of criteria needed to be met before a node was accepted in the answer sample. Regarding stratification, answers were selected from the answer distributions of all five question categories. Regarding randomization, we sampled nodes randomly from the answer distributions of Passage Statements N-2, N-1, N, N+1, and N+2, such that N refers to the queried node. The answer distributions associated with Passage Statement N were weighted higher than those of the other positions (N-2, N-1, N+1, N+2). We began by randomly sampling two nodes from the answer distributions of each of the five question categories associated with Statement N (yielding 10 nodes). Sometimes an answer node was drawn more than once from this initial sample of 10. The other

answer nodes were randomly selected from the answer distributions of Queried Nodes N-2, N-1, N+1, and N+2, again by stratifying the sample across the five question categories. We eliminated nodes from the sample under two conditions. First, a node was eliminated if (a) it was an explicit passage node and (b) 4 explicit nodes had already been selected for that queried node. Second, a node was eliminated if it was a style specification (e.g., X occurred *quickly*) or a time index (e.g., *in the afternoon* and *yesterday*). We focused exclusively on answers that were events, states, actions, and goals because nodes in these categories can be articulated as complete sentences.

It might appear to some readers that the present study is a circular test of the QUEST model. Specifically, the answers to the questions in the present study were extracted from the question-answering protocols in the Graesser and Murachver (1985) study. In fact, however, there was a very low likelihood (6%) that a particular question-answer item in the present study had been generated when a subject was asked the *same* question in the Graesser and Murachver study. That is, any given answer in the present study almost always had been generated by a different question in the Graesser and Murachver study. Therefore, we effectively eliminated the potential circularity in our tests of QUEST.

When the answers were prepared, the events and states were articulated in exactly the same way across the five question categories. Events and states were declarative sentences in the past tense, with no connectives preceding the statement (e.g., "The daughters were frightened." and "It was a nice day."). However, there were fluctuations among question categories as to whether an answer was articulated as an action (e.g., "The dragon kidnapped the daughters.") or a goal (The dragon wanted to kidnap the daughters.). The appropriate forms of articulation were identified for each question category earlier. All goal-action nodes were articulated as actions when they were answers to how-, when-, and CONS-questions. For why-questions, all action-goal nodes within an agent's local goal hierarchy (both superordinate and subordinate) were expressed as goals; all other goal-action nodes were expressed as actions. For enable-questions, the superordinate nodes within an agent's local goal hierarchy were expressed as goals, whereas all other goal-action nodes were expressed as actions. These rules for articulating answers uniformly gave an answer its best shot at being judged as a good answer to a question.

Descriptive statistics were collected on the lexical and linguistic features of the answers. The mean number of syllables per answer was 7.74 ($SD = 2.36$). The mean number of content words (i.e., nouns, verbs, and adjectives) was 3.03 ($SD = .83$). The mean logarithm of word frequency was 2.57 for the content words ($SD = 1.47$).

Apparatus

A microcomputer controlled the presentation of the passages, the presentation of the question-answer items, and the collection of responses. Each subject's station consisted of a sound-shielded booth containing a 12-in. monitor and a keyboard. For those subjects supplying the binary GOA judgments, only two keys could be pressed to advance the screen: a "good" answer key and a "bad" answer key. For those subjects supplying the ratings, there were four keys that corresponded to the following ratings: (a) bad answer to the question, (b) possibly an acceptable answer, (c) moderately good answer, and (d) very good answer.

Procedure

The subjects read one of the two passages two times at their own pace. They were instructed to read the passage for understanding because they would be asked a number of questions about the passage.

The subjects were subsequently presented with several practice question-answer pairs to be sure that they understood the distinction between good answers and bad answers. They were instructed that good answers were not only true statements in the context of the story, but they were "appropriate and relevant" to the question. Bad answers were "inappropriate, irrelevant, or false." Subjects were given feedback on the practice items.

After the practice trials, the subjects provided GOA judgments on the experimental question-answer pairs. Fifty of the subjects furnished discrete GOA judgments, whereas 60 supplied GOA judgments on a four-point scale. There was a separate random ordering of the items for each subject. The subjects began each trial by pressing one of the keys on observing a ready signal. After a .5-s delay, the question appeared on the screen. The subjects read the question at their own pace and pressed a key when finished. After a .5-s delay, the question appeared and the answer appeared on the screen. The subjects provided a GOA judgment by pressing one of the response buttons. The subjects were instructed to make their decisions as soon as possible, but not to sacrifice accuracy and good judgment. Both the responses and the decision latencies were recorded. However, latencies were ultimately not analyzed for GOA judgments on the four-point scale. After the subjects made judgments on all question-answer items in the first story, the procedure was repeated for the second story. The order of the two stories was counterbalanced across subjects.

Predictor Variables in the Multiple Regression Analyses

Each multiple regression analysis contained a single dependent measure (i.e., GOA judgment or decision latency) and several predictor variables. Some predictor variables were theoretically interesting from the perspective of the QUEST model, namely those associated with convergence components and information sources. Operational definitions for the important predictor variables are presented in the introduction. A summary of these predictor variables is provided below.

Arc search procedure. The question-answer item received a score of 1 if there was at least one legal path of arcs between the entry node in the textbase and the answer node. The score was 0 if there was no legal path. The arc search procedure for a question category determined whether there was a legal path between the queried node and the answer node. The arc search procedures differ among the five question categories, so the values on this variable depended on the question category under consideration. The mean arc search scores were .55, .46, .68, .68, and .40 for why-, how-, when-, enable- and CONS-questions, respectively. It should be noted that the textbase (i.e., Figure 1) contained only explicit passage nodes. When the answer was an inference, it was placed at its "virtual location" in the textbase, as if the inference were an explicit statement.

Structural distance. This was the number of arcs between the entry node and the answer node in the textbase. Whenever the answer was an inference, the distance was based on its virtual location, as if the node were in fact stored. Nodes were frequently connected by multiple paths. Structural distance was based on the shortest legal path for legal answers and the shortest path for illegal answers. The mean structural distance score was 1.70 ($SD = 1.24$).

Constraint satisfaction. This variable measured the extent to which the answer satisfied a set of semantic and conceptual constraints imposed by the queried node. The values varied from 0 to 1, as discussed in the Introduction. The values on argument overlap, temporal compatibility, planning compatibility, plausibility, and causal strength were determined by two independent raters who were trained on the five dimensions. The mean reliability scores (Cron-

bach's alpha) for these judgments were quite high: argument overlap (.92), temporal compatibility (.75), planning compatibility (.94), plausibility (.96), and causal strength (.81). The mean values on the dimensions were .61 ($SD = .42$), .54 ($SD = .45$), .94 ($SD = .23$), .92 ($SD = .23$), and .25 ($SD = .32$), respectively. The overall constraint satisfaction score was .65 ($SD = .19$).

Number of information sources. This variable was the number of generic information sources that supplied the answer to the question. Each content word in the queried node served as a potential information source. For example, the information sources for "Why did the heroes fight the dragon?" were generic knowledge structures for *hero*, *fight*, and *dragon*. Decisions needed to be made as to whether a particular answer was stored in each of the information sources. These decisions were based on samples of data collected by Graesser and Clark (1985). Graesser and Clark extracted the content of each generic knowledge structure associated with the two stories. The content was extracted empirically by adopting a "free generation plus question-answering method." Subjects in one group generated lists of properties, actions, events, and other nodes for each information source. Subjects in another group answered why- and how-questions about the content extracted from the free generation sample. A list of statement nodes was prepared for each information source. Each statement node was generated by at least 2 out of 8 subjects in either the free generation task or the question-answering task. Regarding the present study, an answer was scored as coming from an information source if it was a member of the Graesser and Clark node list for that information source. The mean number of generic information sources per answer was .92 ($SD = .92$).

Verbatim statement. This variable specified whether the answer was explicitly mentioned in the passage (value = 1) or whether it was an inference (value = 0). The mean value was .18 ($SD = .38$).

Answer production score. This was the empirical likelihood that the particular answer would be produced when a particular question is asked in a question-answering task. The answer production scores were extracted from the empirical answer distributions collected by Graesser and Murachver (1985). The mean answer production score was .06 ($SD = .15$).

Queried action-event. The queried node was either an action (value = 1) or an event (value = 0). We also originally scaled the items on answer node category. However, answer node category had no significant effects in any of the regression analyses, so it was dropped from the analyses reported in this study.

Story. The Czar story received a value of 1, whereas the Dove story received a value of 0.

An example question-answer item. To illustrate how the question-answer items were scaled, we walk through the measures associated with the question and answer below.

Q: Why did the heroes fight the dragon?

A: Because the heroes wanted to kill the dragon.

The answer is an inference, rather than an explicit text statement, so its virtual location would need to be identified in Figure 1. The location of the goal node would be placed between the queried node (Node 17) and Node 15 ("The heroes wanted to rescue the daughters."). The new node would be connected by a forward Reason arc between the queried node and the new inference node (and a forward Reason arc between the new inference node and Node 15).

Answers to why-action questions include superordinate goals (i.e., those nodes that radiate from the queried node on paths of forward Reason and backward Manner arcs). The inference node would be generated by this arc search procedure for why-action questions, so it would be scored as a legal answer (arc search = 1). The structural distance score would be 1 because the answer is one arc away from the queried node.

Several measures are associated with the constraint satisfaction component. The argument overlap score is 1 because all of the

arguments in the answer (i.e., heroes and dragon) are also in the queried node. The heroes wanting to kill the dragon overlaps temporally with the heroes' fighting the dragon, so the temporal compatibility score is 1. The two goals of fighting and killing are not incompatible, so the planning compatibility score is 1. The plausibility score is 1 because the heroes do want to kill the dragon in the story. The causal strength dimension has four subcriteria. The temporality criterion is satisfied because the goal of killing precedes the action of fighting. The operativity criterion is satisfied because the goal of killing is operative during the action of fighting. The necessity criterion would receive a value of .5 because it is not clear whether the goal of killing is really necessary. The heroes might have fought the dragon merely to harm the dragon or chase it away; the goal of killing could possibly be negated (through the counterfactual test), yet the fighting would occur. The sufficiency criterion would receive a value of 1 because a typical consequence of wanting to kill the dragon would be to fight the dragon. These four scores on the temporality, operativity, necessity, and sufficiency criteria yield an overall causal strength score of .75 [$1 \times 1 \times (.5 + 1)/2$]. The values on the five dimensions of constraint satisfaction yield an overall constraint satisfaction score of .95 ($1 + 1 + 1 + 1 + .75$).

The number of generic information sources that furnish the answer is determined by observing the node lists stored under the relevant generic knowledge structures (GKS). There are three GKSs associated with the queried node: *hero*, *fighting*, and *dragon*. We found that the answer node *X kill Y* was stored under *hero* and *dragon*, but not under *fighting*, when the node lists of Graesser and Clark (1985) were examined. Therefore, this answer would receive the score of 2 for number of information sources.

There are a few additional measures on which the example question-answer item was scaled. The value of verbatim statement is 0 because the answer was not explicitly mentioned in the text. The value of answer production score is .38 because 3 out of 8 subjects produced that answer to "Why did the heroes fight the dragon?" in the Graesser and Murachver (1985) study. The value of queried node category is 1 because the queried node is an intentional action. The value of *story* is 1.

Correlations among predictor variables. Correlation matrices were prepared to assess whether there were any serious problems of colinearity among predictor variables. A correlation matrix was prepared for each of the five question categories. Each matrix included the above predictor variables (with the overall constraint satisfaction measure instead of the breakdown along the five dimensions) and three lexical variables (number of content words, number of syllables, and word frequency). Given that each matrix had 11 variables, there were 55 correlations in each matrix and 95 unique correlations among the five matrices (corresponding to the five question categories). Only 1 out of the 95 correlations was greater than .40; there was a high positive correlation between the number of content words and the number of syllables, $r = .68$, $p < .05$. Another 8 out of the 95 correlations had absolute values of .31 to .40. Therefore, 91% of the correlations were small or modest ($-.30$ to $.30$).

We prepared a second set of five correlation matrices that segregated the five dimensions of constraint satisfaction: argument overlap, temporal compatibility, planning compatibility, plausibility, and causal strength. There were 15 variables in each of these matrices, yielding 105 correlations per matrix and 161 unique correlations among the five matrices. There were only 4 correlations with absolute values of greater than .40. In addition to the high positive correlation between number of syllables and content words, there were positive correlations between causal strength and arc search for why-questions ($r = .45$), enable-questions ($r = .45$), and CONS-questions ($r = .50$). The positive correlation between the causal strength dimension and the arc search procedure was predicted by the QUEST model (see also Trabasso et al., 1989). Another 4 correlations had absolute values

between .31 and .40. Therefore, 95% of the correlations were low or modest ($-.30$ to $.30$).

Results

Multiple regression analyses were conducted to assess the impact of the predictor variables on the discrete GOA judgments, on the GOA ratings, and on the decision latencies for discrete GOA judgments. A beta-weight was associated with each predictor variable, which consisted of a standardized regression coefficient for that variable. Variability among question-answer items (averaging over subjects) served as an error term whenever a multiple regression analysis was performed. The beta-weight reported in the tables and this section are based on these item analyses. However, in all tests of statistical significance, we assessed variability among subjects (i.e., the F_1 statistic) in addition to variability among items (i.e., the F_2 statistic). When variability among subjects was assessed, we performed multiple regression analyses on the data of individual subjects and extracted a beta-weight for each predictor in a subject's analysis. An F test was performed on the subjects' beta-weights for each predictor variable to determine whether the mean beta-weight differed significantly from 0 (see Lorch & Myers, 1990).

We adopted a .05 level in tests of statistical significance, unless specified otherwise. A one-tailed test was adopted whenever QUEST predicted the impact of a variable on the GOA judgments (e.g., the impact of arc search and constraint satisfaction on GOA measures). A two-tailed test was adopted if there was no predicted outcome (e.g., story, queried action-event) or if two hypotheses generated opposite predictions (e.g., structural distance and information sources).

There were three important indices that we considered when we assessed the impact of a predictor variable on a GOA measure. First, we tested whether the predictor variable significantly predicted the GOA measure. Second, we evaluated how consistently the predictor variable affected the GOA measure across the question categories. Third, we observed whether the signs (+ or -) of the beta-weights in the regression equations were in the expected directions. It is important to mention that we never compared the magnitudes of the beta-weights between predictor variables. For example, if the beta-weight for Variable A was .50 and the beta-weight for Variable B was .10, we did not conclude that variable A had a more robust impact on the GOA measure than did variable B.

Goodness-Of-Answer (GOA) Judgments

Means and standard deviations were computed for the discrete GOA judgments and the GOA ratings for the five question categories. The mean discrete GOA judgments for why-, how-, when-, enable-, and CONS-questions were .40, .43, .39, .51, and .50, respectively; the corresponding standard deviations were .34, .29, .31, .33, and .29. The mean GOA ratings were 1.99, 2.07, 2.51, 2.32, and 2.37, whereas the standard deviations were .87, .80, .82, .85, and .70. We were satisfied that these distributions of GOA scores did not present any problem for using multiple regression analyses. The means did not show any ceiling effects or floor effects. The

standard deviations were high enough to rule out a potential problem of there being a restricted range of scores.

Table 3 presents the data from the multiple regression analyses on the discrete GOA judgments and the GOA ratings. Beta-weights are presented for eight predictor variables, segregated by question category. Table 3 presents the proportion of item variance explained by each multiple regression equation. When averaging over the question categories, the multiple regression equations accounted for 50% of the variance of discrete GOA judgments and for 52% of the variance of GOA ratings. All of the multiple regression equations significantly predicted the GOA decisions. Table 3 also indicates whether each predictor variable was significant in the item analysis (signified by *i*) and in the subject analysis (signified by *s*). A predictor variable was declared as statistically significant if it was significant in both the item analysis and the subject analysis.

The regression equations were almost identical for the discrete GOA decisions and the GOA ratings. This can be illustrated by computing the proportion of beta-weights that had the same qualitative outcome (i.e., both were significantly positive, both were significantly negative, or both were non-significant). Of the 40 beta-weight comparisons in Table 3, 35 had the same qualitative outcome (88%). The values of the beta-weights were also similar quantitatively. When the discrete GOA judgments were compared with the GOA ratings, the mean beta-weights (averaging over question category) were virtually identical: arc search (.45 vs. .43, respectively), structural distance (−.09, −.09), constraint satisfaction (.23, .23), information sources (.04, .01), verbatim answer (.04, .06), answer production score (.26, .29), queried action–event (.01, −.01), and story (−.02, .05). There is at least one implication from the fact that the multiple regression equations were virtually the same for the discrete GOA judgments and the four-point GOA ratings. Specifically, the sensitivity of the GOA scale apparently has little or no impact on the outcome of the multiple regression analyses.

Convergence mechanism. Support was found for all three components of the convergence mechanism. Arc search and

constraint satisfaction had significantly positive beta-weights in all 10 multiple regression equations. Structural distance had significantly negative beta-weights in 7 out of 10 equations. The CONS- and when-questions did not show consistent support for structural distance.

Bivariate correlations agreed with the beta-weights presented in Table 3. The GOA judgments were correlated with each predictor variable that was associated with the convergence mechanism. The correlations were perfectly compatible with the beta-weights in Table 3, both in terms of the sign (+ or −) and in terms of statistical significance (significant vs. nonsignificant).

Given that support was found for all three components of QUEST's convergence mechanism, we explored how the three components interacted. A lack of significant interaction among these variables would suggest that the three convergence components are combined independently and additively. On the other hand, there would be statistical interactions among arc search, structural distance, and constraint satisfaction components if the components are executed in a sequential fashion, with one stage (e.g., arc search) preceding another stage (e.g., constraint satisfaction).

We performed multiple regression analyses that assessed interactions among the three components of the convergence mechanism: arc search (A), constraint satisfaction (C), and structural distance (D). This was accomplished by adding four interaction terms to the multiple regression equation: $A \times C$, $A \times D$, $C \times D$, and $A \times C \times D$. The three interaction terms significantly increased the amount of explained variance in the item analyses from 51% to 52%. Two natural groupings of questions emerged on the basis of the patterns of the three-way interaction; why-, how-, and enable-questions formed one group, whereas the when-, and CONS-questions formed the other group. Significant three-way interactions were found for why-questions: discrete GOA, $F_1(1, 9) = 14.06$, $p < .05$, $F_2(1, 307) = 5.48$, $p < .05$, and GOA rating, $F_1(1, 11) = 17.89$, $p < .05$, $F_2(1, 275) = 3.76$, $p < .10$; for how-questions: discrete GOA, $F_1(1, 9) = 1.96$, $p > .05$, $F_2(1, 307) = .94$, $p > .50$, and GOA rating, $F_1(1, 11) = 11.97$, $p < .05$, $F_2(1, 275)$

Table 3
Beta-Weights of Predictors of Goodness-of-Answer (GOA), Segregated by Question Category

Predictor variable	Discrete GOA judgment					GOA rating				
	Why	How	When	Enable	CONS	Why	How	When	Enable	CONS
Convergence mechanism										
Arc search procedure	.49 _{is}	.45 _{is}	.33 _{is}	.51 _{is}	.47 _{is}	.49 _{is}	.45 _{is}	.32 _{is}	.48 _{is}	.41 _{is}
Structural distance	−.16 _{is}	−.13 _{is}	−.06 _s	−.13 _{is}	.03	−.14 _{is}	−.15 _{is}	−.09 _{is}	−.15 _{is}	.06
Constraint satisfaction	.16 _{is}	.31 _{is}	.27 _{is}	.19 _{is}	.24 _{is}	.16 _{is}	.30 _{is}	.35 _{is}	.15 _{is}	.21 _{is}
Information sources										
Generic information sources	.14 _{is}	.02	−.03 _s	.06 _s	−.01	.03 _s	.06 _s	−.05 _s	.05 _s	−.02
Verbatim/inference answer	.01	.10 _{is}	.11 _{is}	.03 _s	.04 _s	.01	.08 _{is}	.16 _{is}	.02	.02
Auxiliary variables										
Answer production score	.29 _{is}	.16 _{is}	.29 _{is}	.22 _{is}	.32 _{is}	.36 _{is}	.20 _{is}	.26 _{is}	.26 _{is}	.35 _{is}
Queried action/event	−.07 _{is}	−.02	.16 _{is}	.02	−.05 _s	−.10 _{is}	−.03 _s	.09 _{is}	.02	−.01
Story	−.08 _{is}	.02	.05	−.07 _s	−.03	−.06 _s	.04	.09 _{is}	.03	.16 _{is}
Variance explained by multiple regression equation (R^2)	.57	.52	.41	.55	.44	.61	.57	.45	.55	.43

Note. *i* = significant at $p < .05$ in item analysis; *s* = significant at $p < .05$ in subject analysis.

= 4.08, $p < .05$; and for enable-questions: discrete GOA, $F_1(1, 9) = 24.60$, $p < .05$, $F_2(1, 307) = 4.20$, $p < .05$, and GOA ratings, $F_1(1, 11) = 17.64$, $p < .05$, $F_2(1, 275) = 3.92$, $p < .05$. The patterns of these three-way interactions were the same for why-, how-, and enable-questions, as is discussed shortly. In contrast, significant three-way interactions did not emerge when we analyzed when- and CONS-questions.

Figure 2 plots the three-way interactions for the two groups of questions, segregating the discrete GOA judgments and the GOA ratings. The *b-weights* (i.e., nonstandardized regression coefficients) of the three main effects and the four interaction terms were used to generate the values in Figure 2. The values of arc search were either 0 (illegal) or 1 (legal); the values of constraint satisfaction were 0 (satisfy no constraints) or 1 (satisfy all constraints); the values of structural distance were 1, 2, or 3. As shown in Figure 2, structural distance consistently showed flat lines for when- and CONS-questions. We confirmed this by computing bivariate correlations between

structural distance and GOA on four subsets of items: legal answers with high constraint satisfaction scores (using a median split criterion), legal answers with low constraint satisfaction scores, illegal answers with high constraint satisfaction scores, and illegal answers with low constraint satisfaction scores. Only 1 of these 16 correlations between structural distance and GOA was statistically significant; the mean correlation was $r = -.07$. Structural distance clearly had no impact on the GOA judgments for when- and CONS-questions.

In the case of why-, how-, and enable-questions, however, structural distance significantly affected GOA decisions for those legal answers that failed to satisfy constraints. The other three lines within each group in Figure 2 were essentially flat. Structural distance was a criterion for evaluating GOA only when the answer passed the arc search procedure, but failed the constraint satisfaction component. Bivariate correlations between structural distance and GOA decisions were consist-

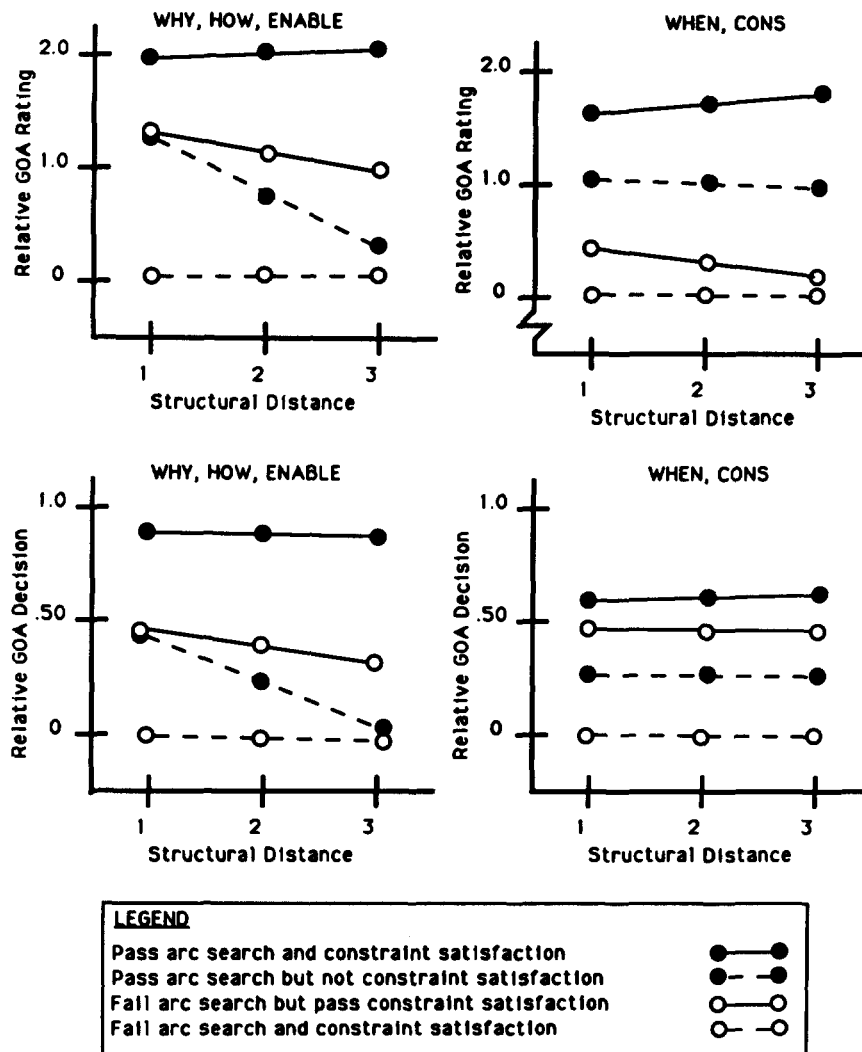


Figure 2. Three-way interaction among arc search, constraint satisfaction, and structural distance, segregated by question category and goodness-of-answer (GOA) scale. (CONS = consequence.)

ent with the lines in Figure 2. These correlations were consistently high and significant for legal answers with low constraint satisfaction scores; six out of six correlations were significantly negative, with a mean correlation of $r = .42$. In contrast, the bivariate correlations were low and not consistently significant for the other three categories of answers (mean correlation of $-.03$).

The results failed to support a sequential model that assumes that the arc search procedure is executed before constraint satisfaction. If anything, the arc search and constraint satisfaction components are executed simultaneously, in an additive fashion. Both legal and illegal answers showed robust effects of constraint satisfaction; a sequential model predicts that constraint satisfaction would influence GOA judgments for legal answers, but not for illegal answers. The $A \times C$ interaction term was nonsignificant in 7 out of 10 multiple regression equations, an outcome that would support an additive model. The $A \times C$ interaction was significant for why-questions (both discrete judgments and ratings) and enable-questions (ratings only). The pattern of these three significant interactions indicated that constraint satisfaction had a more robust effect on legal answers than on illegal answers. Nevertheless, analyses of all question categories indicated that GOA judgments were significantly higher for illegal answers that satisfy constraints than for illegal answers that fail to satisfy constraints. Once again, these data consistently argue against a sequential model in which the arc search procedure is executed before the constraint satisfaction component.

There was support for the claim that the arc search and constraint satisfaction components precede the evaluation of structural distance, at least for why-, how-, and enable-questions. This claim is supported by the three-way interaction plotted in Figure 2. The data support a processing model that is depicted in Figure 3. After the question and answer is interpreted, the arc search and constraint satisfaction components are evaluated independently, perhaps in parallel. The

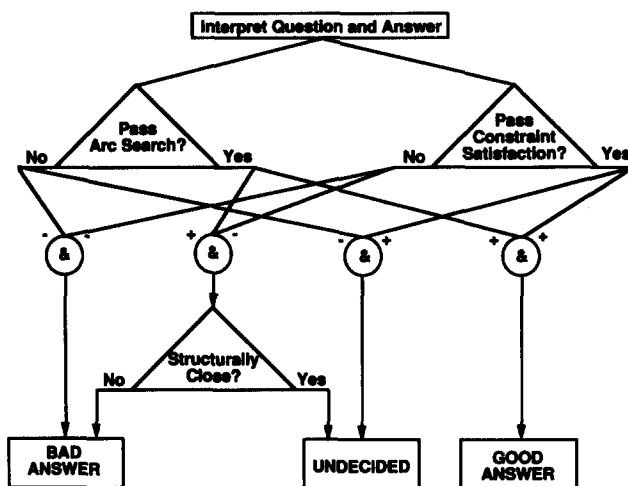


Figure 3. A model that specifies how arc search, structural distance, and constraint satisfaction interact in the case of why-, how-, and enable-questions.

outcome of each component is either positive (good answer) or negative (bad answer). Structural distance is computed in only one condition, namely when the output of arc search is positive and the output of constraint satisfaction is negative. In all other conditions, structural distance has no significant impact on GOA decisions. Moreover, structural distance has no impact on GOA decisions for when- and CONS-questions. For these question categories, the "structurally close" decision component is entirely bypassed in the case of legal answers that fail constraints; these items are linked directly to the "undecided" alternative.

Dimensions of constraint satisfaction. We performed a set of multiple regression analyses that segregated the five dimensions of constraint satisfaction. The amount of explained item variance increased from 51% to 55% when we segregated the five dimensions. When averaging across the five question categories and two GOA scales, the mean beta-weights were $-.02$, $.10$, $.03$, $.14$, and $.19$ for argument overlap, temporal compatibility, planning compatibility, plausibility, and causal strength, respectively. Causal strength was significant in all 10 analyses, temporal compatibility and plausibility were each significant in 7 analyses, planning compatibility was significant in 4 analyses, and argument overlap was significant in 2 analyses. The two significant effects of the argument overlap variable were opposite in sign (+ in one analysis and - in the other), so there was essentially no support for argument overlap. Nearly all of the significant predictors (29 out of 30) were in the direction that would be predicted by QUEST. In summary, there was evidence for four out of the five dimensions of constraint satisfaction.

Answers with answer production scores of zero. Multiple regression analyses were performed on those answers that had answer production scores of 0. These answers to a particular question were never generated by subjects who supplied question-answering protocols in the Graesser and Murachver (1985) study. The same 10 multiple regression equations were performed as those in Table 3, except that answer production score was not included as a predictor variable (because the value was always 0). All 10 regression equations significantly predicted the GOA judgments, accounting for 39% of the item variance overall. The beta-weights were qualitatively and quantitatively similar to those in Table 3. Of the 70 beta-weights in these analyses, 65 had the same qualitative outcome as those in Table 2; that is, they were both significantly positive, both significantly negative, or both nonsignificant. The mean beta-weights of the theoretically interesting predictors were quantitatively similar for (a) the entire answer sample and (b) those answers with answer production scores of 0: arc search ($.44$ and $.44$, respectively), constraint satisfaction ($.23$, $.28$), structural distance ($-.09$, $-.09$), information sources ($.03$, $.02$), and verbatim answer ($.05$, $.04$).

Alternative measures of structural distance. Structural distance was measured in the previous analyses by counting the number of arcs between the queried node and the answer node in the explicit textbase. This measure of structural distance was successful in predicting GOA judgments in the previous regression analyses. The bivariate correlation between structural distance and GOA judgments was significant

in three out of five question categories (mean $r = -.24$); the exceptions were the when- and CONS-questions.

It is quite possible that the textbase contains inferences in addition to the explicit passage statements. If this is true, then a better measure of structural distance would be derived from a textbase that contained inferences. We first considered the possibility that the textbase had a large volume of inferences. For each story, we constructed a large textbase that contained (a) the explicit passage statements and (b) all unique inferences produced by why-, how-, when-, enable-, and CONS-questions in the Graesser and Murachver (1985) study. There were over 200 statement nodes in the resulting textbase of each story. This revised textbase with all inferences provided a poorer measure of structural distance than did the explicit textbase. The mean correlation was only $-.10$ when structural distance from the revised textbase was correlated with GOA judgments. This correlation was significantly lower than the $-.24$ correlation associated with the explicit textbase.

It is possible that the textbase contains some, but not all, of the inferences in the Graesser and Murachver sample (1985). According to one hypothesis, those inferences elicited by why-questions tend to be comprehension-generated and incorporated into the textbase, whereas the inferences elicited by the other types of questions tend not to be comprehension-generated (Graesser & Clark, 1985; Long et al., 1990). Therefore, we constructed a textbase that included explicit passage statements and inferences generated from why-questions. The revised textbase from each story contained approximately 100 statement nodes. Once again, however, the revised textbase did not fare as well as the explicit textbase. The structural distance scores from the revised textbase had a $-.14$ correlation with GOA judgments, which is lower than the $-.24$ correlation associated with the explicit textbase. The conclusion from these analyses is that the explicit textbase provided the best measure of structural distance.

When the pragmatic components were discussed in the Introduction, there was some foundation for expecting a curvilinear relationship between structural distance and GOA judgments. That is, there would be an inverted-U function, such that answers at intermediate distances would be the best answers. To test this possibility, multiple regression analyses were performed with two predictors, namely structural distance (D) and D^2 . Only 1 of the 10 multiple regression analyses confirmed the inverted-U function by showing a significant negative D^2 term. Therefore, there was no consistent support for a curvilinear relationship between structural distance and GOA judgments.

Information sources. The multiple regression analyses failed to support the prediction that GOA judgments would increase as a function of the number of information sources. The mean beta-weight among the 10 multiple regression analyses was only .03; only 1 out of the 10 analyses showed a significant effect of number of information sources. Only 3 out of the 10 bivariate correlations were statistically significant (mean $r = .11$).

There is some foundation for anticipating an inverted-U function between number of information sources and GOA judgments. Specifically, answers that come from many information sources are comparatively uninformative, whereas

answers that come from no information sources are difficult to interpret. To assess whether there was a significant inverted-U function, we performed 10 regression analyses with two predictors: Number of information sources (I) and I^2 . A significantly negative I^2 term would occur if there was an inverted-U function. However, the I^2 term was not significantly negative in any of the 10 analyses. When we restricted our analyses to legal answers (passing the arc search procedure), once again there were no significantly negative I^2 terms. Therefore, there was no evidence for the potential curvilinear relationship between number of information sources and GOA judgments.

Multiple regression analyses were conducted to assess whether there were significant interactions between number of information sources and the components of convergence (arc search, structural distance, and constraint satisfaction). Thirty multiple regression analyses were conducted: 2 (discrete judgment vs. rating) $\times$ 5 (question categories) $\times$ 3 ($I \times A$ and $I \times C$ vs. $I \times D$ interactions). Only 2 of the 30 analyses showed a significant interaction, so there was no consistent support for the possibility that number of information sources interacts with the convergence mechanism.

One of the information sources is the explicit textbase. The verbatim nodes were directly stored in the textbase, whereas the inferences were presumably constructed when the GOA judgment was made. How- and when-questions showed a significant advantage for verbatim answers over inference answers (mean beta-weight = .11), whereas there was no such advantage for why-, enable-, and CONS-questions (mean beta-weight = .01).

Decision Latencies

Decision latencies were collected on the discrete GOA judgments. Mean decision latencies for why-, how-, when-, enable-, and CONS-questions were 2.39, 2.40, 2.79, 2.51, and 2.69 s, respectively. The corresponding standard deviations were .58, .58, .60, .54, and .61.

All analyses of decision latencies included number of content words, number of syllables, and word frequency as predictor variables in addition to the eight predictors in Table 3. Therefore, when main effects were analyzed, each multiple regression equation had 11 predictor variables.

Table 4 presents the beta-weights from the multiple regression analyses, segregated by the five question categories. Separate analyses are presented on the complete set of 320 question-answer items and on those items with answer production scores of 0. Each of the 10 multiple regression analyses were statistically significant. The mean amount of decision latency variance explained was 35%. The beta-weights were quantitatively similar between the 320-item sample and the sample with answer production scores of 0: arc search (.04 vs. .08, respectively), structural distance ($-.04$, $-.03$), constraint satisfaction (.02, .00), information sources (.04, .07), verbatim statement ($-.07$, $-.07$), queried event-action (.00, .00), story (.17, .13), content words (.22, .22), syllables (.28, .28), and word frequency (.00, .00).

Components of convergence mechanism. According to the beta-weights in Table 4, legal answers to why-, how-, and

Table 4
Beta-Weights of Predictors of Goodness-of-Answer (GOA) Decision Latencies

Predictor variable	All Q/A items					Q/A items with answer production scores of 0				
	Why	How	When	Enable	CONS	Why	How	When	Enable	CONS
Convergence mechanism										
Arc search procedure	.16 _{is}	.09 _{is}	.02	.10	-.15 _{is}	.24 _{is}	.06 _s	.09	.12	-.10 _s
Structural distance	-.05 _s	-.07 _s	-.04	-.07	.05	-.05 _s	-.08 _s	-.03	-.06	.05
Constraint satisfaction	.10 _{is}	.01	.01	-.02	.01	.08 _s	-.04	.04 _s	-.04	-.05
Information sources										
Generic information sources	.03	.11 _i	.05	.00	.01	.12 _i	.07	.16 _i	.03	-.03
Verbatim/inference answer	-.02	-.14 _{is}	-.06	-.09 _i	-.04	-.02	-.14 _{is}	-.06	-.10 _i	-.04 _s
Auxiliary variables										
Answer production score	-.27 _{is}	-.22 _i	-.17 _{is}	-.16 _{is}	-.25 _{is}	—	—	—	—	—
Queried action/event	-.07	-.03	.03	.05 _s	.04	-.06	-.03	.04	.04 _s	.03
Story	.07	.25 _i	.14 _i	.16 _i	.23 _{is}	.05	.18	.19	.10	.14
Content words	.36 _{is}	.19 _{is}	.26 _{is}	.17 _{is}	.16 _{is}	.36 _{is}	.20 _{is}	.26 _{is}	.18 _{is}	.17 _{is}
Syllables	.24 _{is}	.35 _{is}	.20 _{is}	.29 _{is}	.30 _{is}	.24 _{is}	.35 _{is}	.21 _{is}	.28 _{is}	.30 _{is}
Word frequency	.02	.00	.05	-.06	.00	.02	.02	.04	-.07	.00
Variance explained by multiple regression equation (R ²)	.42	.38	.29	.26	.38	.43	.38	.30	.27	.38

Note. Q/A = question-answer; i = significant at $p < .05$ in item analysis; s = significant at $p < .05$ in subject analysis.

enable-questions tended to have longer decision times than did the illegal answers; three out of the six beta-weights for arc search were significantly positive in Table 4. In contrast, these beta-weights for when- and CONS-questions were either nonsignificant or significantly negative. The beta-weights for structural distance were negative in 8 out of 10 analyses, but were never significant. The beta-weights for constraint satisfaction were significant in only 1 out of 10 analyses.

We performed follow-up multiple regression analyses that assessed interactions among the three components of convergence ($A \times C$, $A \times D$, $C \times D$, and $A \times C \times D$). The three-way interaction was significant in only 2 out of the 10 analyses; the two significant three-way interactions did not have similar patterns and were difficult to interpret. Therefore, we performed analyses on two-way interactions. Structural distance did not significantly interact with the other two variables in any of the analyses with two-way interactions. We performed a set of analyses on the $A \times C$ interaction term along with the other 11 predictor variables in Table 4. This $A \times C$ interaction term was either significant or almost significant for why-questions, $F_1(1, 9) = 3.31$, $p = .10$, $F_2(1, 307) = 3.24$, $p < .10$; for how-questions, $F_1(1, 9) = 2.34$, $p < .20$, $F_2(1, 307) = 5.02$, $p < .05$; and for enable-questions, $F_1(1, 9) = 6.70$, $p < .05$, $F_2(1, 307) = 2.72$, $p < .10$, but was not significant for when- and CONS-questions (all $F_s < 1$). It should be noted that the GOA judgments also manifested natural groupings for why-, how-, and enable-questions versus when- and CONS-questions.

Figure 4 plots the interaction between arc search and constraint satisfaction, segregating the why-, how-, and enable-questions from the when- and CONS-questions. Each interaction was generated on the basis of the b-weights associated with A, C, and $A \times C$ predictors in the equation. For when- and CONS-questions, arc search and constraint satisfaction had essentially no effects on decision latencies. In contrast,

there was an interesting pattern of latencies for why-, how-, and enable-questions. The pattern indeed provided further support for the processing mechanism in Figure 3. According to Figure 3, the only answers that involved an evaluation of structural distance were those that passed arc search, but failed constraint satisfaction components. This prediction was in fact confirmed. Statistical tests revealed the following pattern of latencies: legal answers that fail constraints > illegal answers that pass constraints = legal answers that pass constraints > illegal answers that fail constraints. Latencies tended to be longest when there was a discrepancy between the arc search component and constraint satisfaction components; these two components were apparently evaluated in parallel and independently. One way to resolve the discrepancy and to break

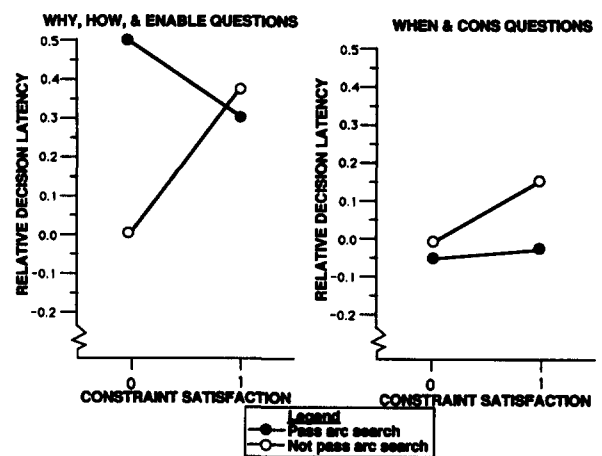


Figure 4. Relative GOA latencies as a function of arc search and constraint satisfaction components, segregated by question categories. (CONS = consequence.)

the tie was to compute the structural distance between the queried node and answer node. However, additional time was needed to compute structural distance.

Information sources. Decision latencies were not affected by the number of generic information sources (mean beta = .04). We performed some follow-up multiple regression analyses that assessed interactions between the information source variable and the three convergence components ($I \times A$, $I \times C$, and $I \times D$). Only 2 out of 15 possible interactions were statistically significant. The impact of number of generic information sources on decision times appears to be either unsystematic or negligible. According to Table 4, the decision latencies were somewhat faster for verbatim answers that are stored in the textbase than for inference answers because all of the beta-weights were negative (mean beta = -.07). However, only the how-questions showed a significant advantage for verbatim answers over inferences. In summary, the decision latencies were not significantly influenced by either the generic information sources or the specific textbase.

Other predictor variables. The beta-weights were in the expected direction for answer production scores (mean beta = -.21), content words (mean beta = .22), and syllables (mean beta = .28). Word frequency and queried action-event did not influence decision latencies (mean beta = .00).

Discussion

All three components of QUEST's convergence mechanism were supported in our analyses of GOA judgments. GOA judgments were higher for legal answers that passed the arc search procedure than for illegal answers. GOA scores were higher for answers that were conceptually compatible with the queried node than for answers that failed to satisfy these constraints. GOA scores decreased as a function of structural distance when the why-, how-, and enable-questions were analyzed, but there was no such relationship for when- and CONS-questions.

Previous studies (Golding et al., 1990; Murachver et al., 1985) reported that GOA judgments are significantly predicted by the arc search and structural distance components when answers are evaluated in the context of narrative text. Arc search and structural distance are also robust predictors of answer production scores when question-answering protocols (rather than GOA judgments) are collected in the context of narrative text (Graesser & Murachver, 1985) and expository text (Graesser & Hemphill, 1991; Graesser, Hemphill, & Brainerd, 1989). The present study is the first to present evidence for the constraint satisfaction component. As expected, GOA scores increase to the extent that the answers satisfy more constraints.

The patterns of GOA judgments and decision latencies provide some clues about the interaction among the three convergence components. The processing of why-, how-, and enable-questions were somewhat different from that of the when- and CONS-questions. Consider first the why-, how-, and enable-questions. Figure 3 presents a model that shows how the arc search, constraint satisfaction, and structure distance components interact for these three question categories. The arc search and constraint satisfaction components

are executed in parallel and independently, rather than sequentially. If arc search had preceded constraint satisfaction in a sequential model, then constraint satisfaction would have influenced the GOA judgments for legal answers (i.e., those that pass the arc search procedure), but not for illegal answers. However, the data indicated that constraint satisfaction had approximately the same impact on both legal and illegal answers; such a finding supports a model in which these two components are executed independently (as in Figure 3).

In some cases, the output from arc search and constraint satisfaction components are in agreement (i.e., both good or both bad answers) and the appropriate GOA decision is made. The decision latencies indicated that more time is needed to determine that the answer is more good than it is bad; good answers require that multiple criteria be satisfied, whereas bad answers can be identified as soon as there is a failure on a few criteria. In other cases, there is a discrepancy in the output from the arc search and constraint satisfaction components. When the output from the arc search component is positive, but the output from the constraint satisfaction component is negative, then additional time is needed to evaluate the answer on structural distance. For these answers, an answer that is structurally close to the queried node would be considered moderately good, whereas an answer is bad when it is many arcs away from the queried node. Legal answers that failed constraint satisfaction were the only items that showed a significant negative relationship between GOA judgments and structural distance. Subjects apparently used structural distance as a criterion for resolving the conflict between the arc search procedure and the constraint satisfaction components. When the answer fails arc search, but passes constraint satisfaction, the subject is undecided, and the decision latency is moderately long.

The structural distance evaluator does *not* consist of a process of searching through a structure in the form of spreading activation or marker passing. If that had been the case, there would have been longer latencies for those answers that were many arcs from the queried node. If anything, the latencies were faster for answers that were at a greater distance from the queried node; such an outcome is clearly incompatible with any search mechanism that involves spreading activation or marker passing. The structural distance evaluator appears to involve a process of discriminating whether the answer is close to the queried node in the text. Alternatively, it involves discriminating whether the answer node would occur before or after the queried node within the story plot. Such discriminations are accomplished more quickly when the answer is more distant from the queried node, as would be the case in any comparative judgment task (Banks, 1977; Graesser & Hemphill, 1991; Wagener-Wender & Wender, 1990).

Structural distance did not play a significant role when individuals judged answers to when- and CONS-questions. In essence, the structural distance component in Figure 3 is absent for when- and CONS-questions. Graesser and Murachver (1985) also reported that structural distance plays a minimal role for CONS- and when-questions in question-answering tasks. They observed that good answers to CONS-questions consisted of important actions-events in the plot, regardless of the structural distance between the queried node

and answer node. Thus, an episode in the plot can have repercussions on subsequent episodes that extend far into the future. In both the present study and the Graesser and Murachver study, it was found that answers to when-questions tend to be verbatim statements in the text, rather than inferences, with no impact of structural distance. The fact that structural distance failed to have an impact on when-questions suggests that the answerer is merely making temporal judgments as to whether the queried event or the answer occurred first in the plot, without worrying whether the answer was tied locally to the queried node in the textbase.

Once again, for all question categories, there was no evidence for a sequential order in the processing of arc search and constraint satisfaction. The constraint satisfaction component was significant both for legal and illegal answers; these two components combined in an additive fashion. If arc search had preceded constraint satisfaction, then there would have been a significant interaction between these two components, with constraint satisfaction having a small impact on illegal answers and a large impact on legal answers. In contrast, there was evidence for sequential processing when structural distance was analyzed. Structural distance is evaluated after the arc search and constraint satisfaction components are completed, at least for why-, how-, and enable-questions.

Arc Search Procedures

QUEST's arc search procedures have robustly predicted answer production scores and GOA judgments in the context of both narrative and expository text. Although these arc search procedures significantly discriminate good from bad answers, some of these procedures could be better tuned to determine exactly what combinations of arc categories (and directions) are associated with particular question categories.

We were reasonably satisfied with the arc search procedures of why- and how-questions. Answers to why-event questions are on paths of causal antecedents to the queried event (combinations of Implies arcs and backward C, O, and I arcs) and are directly analogous to the "causal antecedent" questions in Lehnert's QUALM model (Lehnert, 1978). Answers to how-event questions include both causal antecedents and nodes that specify the manner or style in which the event sequence unfolds (through forward M arcs). Answers to why-action questions sample superordinate goals in the local goal hierarchy of the agent (through forward R and backward M arcs) as well as causal antecedents to the local goal hierarchy. This arc search procedure corresponds to the "goal-orientation" question category in Lehnert's QUALM model. Answers to how-action questions sample subordinate actions in an agent's local goal hierarchy (through backward R and forward M arcs), causal antecedents to the local goal hierarchy, and style embellishers (through forward M arcs). The arc search procedures for how-action questions in QUEST are analogous to Lehnert's "instrumental-procedural" questions. In summary, the arc search procedures for why- and how-questions are not only supported by empirical data, but are also consistent with models of question answering in artificial intelli-

gence (Lehnert, 1978; Lehnert et al., 1983; Souther et al., 1989; Winston, 1984).

We are not entirely satisfied with QUEST's arc search procedures for when-, enable-, and CONS-questions. First, the arc search component did not predict GOA judgments for when- and CONS-questions as well as it predicted the judgments for the other three question categories. Second, illegal answers to when-event questions might be acceptable if the right connective precedes the answer. In particular, the arc search procedure for when-event questions generates causal antecedents, but not causal consequences; subjects rarely generate causal consequences as answers to when-event questions in the context of stories and expository text (Graesser & Hemphill, 1991; Graesser, Hemphill, & Brainerd, 1989; Graesser & Murachver, 1985). However, causal consequences are acceptable if the answer is preceded by the connective *before*: "When were the daughters frightened? Before the heroes heard their cries." One direction for future research is to tune the arc search procedures and to examine how these procedures are influenced by connectives (e.g., *because*, *in order to*, *before*, *after*, and *during*).

Constraint Satisfaction

Constraint satisfaction is essentially a process of answer node evaluation. That is, the subject evaluates whether the conceptual properties of the answer nodes are correct and compatible with the queried node. Four out of the five dimensions of constraint satisfaction ended up predicting GOA judgments in the present study. Argument overlap was the only dimension that failed to predict GOA judgments.

There was uniform support for the causal strength dimension, which was based on Trabasso and van den Broek's analysis of causality in narrative text (Trabasso & van den Broek, 1985; Trabasso et al., 1988, 1989; van den Broek, 1990). We should point out that the causal strength dimension is not strictly equivalent to the Consequence arc in a causal network. The Consequence arc is a weaker form of causality than the causal strength dimension. Causal strength is evaluated on the basis of temporality, operativity, necessity, and sufficiency, whereas a Consequence arc is evaluated on the basis of temporality and the "because test" (i.e., it is more appropriate to say "B because A" than to say "A because B"). Structures frequently have chains of nodes that are connected by Consequence arcs: (X) -C→ (Y) -C→ (Z). Indeed, there normally is a thread of a causal chain that spans the entire plot of a short story. The arc search procedure traverses these Consequence arcs as if the Consequence relation were transitive. However, transitivity is not guaranteed when causal strength is evaluated as a component of constraint satisfaction. That is, X might have high causal strength with Y, and Y might have high causal strength with Z, yet X could have a very low causal strength with Z. Although causal strength is not strictly equivalent to the Consequence arc, there is some correlation between the causal strength dimension and the arc search procedures that traverse causal chains.

The dimension of temporal compatibility determines whether the duration of the answer node falls within the time span of the queried node. For example, if the queried node is

"the daughters cried," then "the daughters left the palace" would be outside of the time span of their crying, whereas "the daughters were frightened" would be within the time span. There have been a few models in artificial intelligence that represent time as duration spans (i.e., time intervals, rather than points along a time line) and that generate temporal inferences by propagating constraints according to a time interval algebra (Allen, 1984; Vilain & Kautz, 1986). There are two potential problems with the temporal compatibility dimension that merits attention in the future. First, there is some uncertainty about the most appropriate way to make the judgment of temporal compatibility in some situations. When the queried node is an action (e.g., the heroes fought the dragon), there is some uncertainty about whether to compute the duration of the action per se or to compute the duration of the local plan of the agent (e.g., going to the dragon, rescuing the daughters, and returning the daughters to the palace). Second, it might not be appropriate to apply the temporal compatibility constraint to some question categories. An answer to a CONS-question might be perfectly appropriate, but not within the time span of the queried node "What are the consequences of the heroes fighting the dragon? The Czar rewarded the heroes."

According to the constraint of planning compatibility, the answer node must not interfere with the plan of the queried action. Some computational theories of story plot construction have emphasized the role of agents' plans and plan interactions (Alterman, 1988; Dyer, 1983; Reiser, Black, & Lehnert, 1985; Wilensky, 1983). These theories inspired our inclusion of planning compatibility as a constraint in the QUEST model. It should be noted, however, that this dimension might operate quite differently for some question categories. For example, a good answer might directly violate the plan of the queried node in the case of some question categories, such as *why-not-action* questions (e.g., Why didn't the dragon kill the daughters? The dragon wanted to play with the daughters.).

The constraints of plausibility (truth) states that the answer cannot be a false statement from the standpoint of the knowledge structures under consideration. Obviously, the answer should not contradict the queried node or another stored fact in the relevant knowledge base. There are models of human plausible reasoning that specify how world knowledge and reasoning heuristics together compute the truth or plausibility of a statement, even when the statement is not directly stored in the knowledge base (Collins & Michalsky, 1989). Reder (1982, 1987) reported that generic knowledge may be used to evaluate the plausibility of a statement even when that statement was explicitly mentioned in the past and could be directly retrieved from memory.

Information Sources

According to QUEST, the information sources include the textbase and the generic knowledge structures that are activated by the content words in the queried node. Regarding the textbase, we found that the explicit textbase provided a better metric of structural distance than did (a) a textbase that contained a massive volume of knowledge-based inferences

and (b) a textbase that included a subset of inferences that are needed for establishing causal coherence and inferences that are otherwise generated on-line (i.e., answers to why-questions; Long et al., 1990). This outcome is compatible with theories of text representation that segregate an explicit textbase (containing very few inferences) from a situation model that houses world knowledge (Kintsch, 1988; Kintsch & van Dijk, 1978; McKoon & Ratcliff, 1989).

When an answer was an inference in the present experiment, the inference was fit into the explicit textbase at its "virtual location." However, the verbatim answers did not necessarily receive higher GOA scores than did the inferences. Verbatim answers were somewhat better answers than were inferences in the case of how- and when-questions, but not in the case of why-, enable-, and CONS-questions.

The number of generic information sources activated by a question had a negligible impact on GOA judgments and latencies. We expected that the number of information sources would have a positive correlation with GOA if the pragmatic component of QUEST had no impact on this experiment. If pragmatics did play an important role, however, then there would be either (a) a negative relationship, because answers from multiple information sources are uninformative, or (b) an inverted-U curvilinear relationship. However, none of these alternatives was confirmed. One reason for the small impact of multiple information sources on GOA judgments may be that the textbase involved simple stories with familiar content words. The content words triggered generic knowledge structures that were well-learned and highly automatized. Perhaps expository texts on difficult, unfamiliar topics would show greater effects of multiple information sources.

According to Reder (1982, 1987), individuals can strategically utilize either the textbase or the generic information sources when they provide plausibility judgments and recognition judgments. The present study was not designed to test this claim directly and we are uncertain about the extent to which the present tasks utilized the two classes of information sources. The fact that the explicit textbase provided the best metric of structural distance would perhaps suggest that the textbase prevailed; yet the small and inconsistent effect of the verbatim-inference contrast suggests that the generic knowledge prevailed. The fact that the question-answer task occurred under immediate testing conditions (rather than a delay of one day or longer) would shift the balance toward the textbase prevailing. In contrast, the fact that there was massive output interference (i.e., 320 question-answer test items, with many nodes being repeated) would perhaps shift the balance toward generic knowledge.

The QUEST model was tested in the context of only one theory of knowledge representation, namely the conceptual graph structures adopted by Graesser (Graesser & Clark, 1985; Graesser & Franklin, 1990). It is important to develop and test the arc search procedures in the context of other theories of knowledge representation, such as conceptual dependency theory (Schank & Abelson, 1977) and Trabasso's causal chain theory (Trabasso et al., 1988, 1989). In the best of all possible worlds, there is an optimal match between the arc search procedures associated with various question categories and a

theory of knowledge representation. One direction for subsequent research is to compare alternative theories of knowledge representation with respect to their ability to explain question-answering data.

Pragmatics

The present study was not designed to critically test the pragmatic components of question answering. We did, however, consider a few hypotheses that would be generated by pragmatics. In particular, the common ground between questioner and answerer might have influenced the GOA judgments. Those answers that are furnished by multiple information sources are presumably in the common ground and comparatively uninformative, which would yield lower GOA judgments. If so, GOA judgments would show either a negative correlation or an inverted-U relationship with number of information sources. We found no evidence for such trends. Similarly, there was some foundation for expecting an inverted-U relationship between GOA scores and structural distance, but such a trend was not statistically significant. It appears that the pragmatic components play a minimal role in the experimental context of having subjects give GOA judgments on computers in the context of simple stories. To provide more informative tests of the pragmatic components, question answering would need to be investigated in more naturalistic settings in which the questioner and answerer have genuine agenda (see Graesser, Roberts, & Hackett-Renner, 1990; van der Meij, 1987).

Graesser, Roberts, and Hackett-Renner (1990) reported three studies that examined the extent to which QUEST's arc search procedures could account for answers to why-, how-, when-, and CONS-questions in naturalistic contexts. In one study the context was a telephone survey in which a question was asked about a historical event (e.g., *How did the Titanic sink?*). In a second study college students visited businesses in the guise of customers, asked clerks questions (e.g., "How does a person get a credit card?" in a bank), and tape-recorded answers. A third study was an analysis of question answering in television interviews (McNeil/Lehrer News Hour) and science films. QUEST's arc search procedures could account for 83% of the answers to the questions and 94% of the answers that referred to the topic being queried (e.g., the sinking of the Titanic and acquiring a credit card). The answers that were outside of the scope of QUEST included counterquestions, directives, requests, expressive evaluations, and other speech acts that addressed the pragmatic interaction between questioner and answerer. These data suggest that the arc search procedures of the QUEST model have impressive external validity.

In closing, we believe that the QUEST model provides a solid beginning for developing a general psychological theory of question answering. However, QUEST needs to be tested in the context of a broader array of texts, social situations, and conversations.

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Received June 12, 1990

Revision received January 15, 1991

Accepted January 22, 1991 ■